

UNDERGRADUATE THESIS

Bachelor's Degree in Data Science for Business

Purchasing Behavior in a Uruguayan Supermarket Chain

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EXECUTIVE SUMMARY

The supermarket chain analyzed is one of the retail chains with the largest presence in Uruguay and one of the most relevant players in the supermarket sector. Its loyalty program generates a very significant volume of transactional information, which opens up multiple opportunities for analysis, model development and business understanding. In this context, the goal of this project was to transform this massive database into structured, analytically valuable information, capable of offering a first approximation of customers' purchasing behavior and supporting strategic decision-making.

First, a comprehensive analysis of the loyalty program data was carried out, including cleaning, integration and a detailed exploration of consumption behavior. Second, a demonstrative clustering was implemented, based on vector representations of products obtained through Word2Vec and the K-Means algorithm. This approach made it possible to condense the complexity of the product universe into numerical vectors and, from them, identify subtle but consistent differences in purchasing patterns across different groups of customers.

Finally, an interactive Power BI dashboard connected to the corporate database was built, which complements the previous analysis by offering a clear and up-to-date view of the chain's aggregate demand. This tool makes it possible to observe the evolution of sales across different time scales, identify seasonality, compare performance across stores, subclasses and suppliers, and analyze how demand varies by store format and sales channel. Its structure facilitates continuous monitoring and helps support operational decisions related to commercial planning, assortment management and work with suppliers.

The results represent a significant contribution for the company. On the one hand, they make it possible to move from a purely transactional view to a more structured understanding of customer behavior. On the other hand, they provide inputs that can be used to rethink the loyalty program, design incentives better aligned with actual consumption habits, identify commercial opportunities and improve the business's analytical capabilities. If the data scope and computing infrastructure were expanded, these techniques could evolve into more precise segmentations and predictive models of great strategic value.

1. SCOPE

SCOPE

The main scope of this project is the analysis of loyalty-program customers of a Uruguayan supermarket chain, using the data available in its loyalty program. The study focuses on customers' purchasing behavior, incorporating demographic variables (age, gender and socioeconomic level), transactional variables (purchase frequency, volume, ticket amount) and product variables (categories, product lines, brands and combinations purchased).

The analysis is developed from two complementary dimensions:

- Transactional perspective, based on purchase tickets as the unit of analysis. This approach makes it possible to study consumption behavior in aggregate, considering each ticket as a manifestation of the act of purchase and, taken together, as a reflection of customers' general behavior patterns. In this way, the aim is to identify groups of tickets with similar consumption characteristics, representing different types of purchasing behavior within the company's ecosystem.
- Temporal perspective, aimed at detecting seasonality and trends over time, analyzing the evolution of consumption and possible changes in customer preferences.

The work is limited to the supermarket business unit, since it is the area where data collection and systematization are most advanced and where the company's strategic interest is concentrated. Other business units belonging to the corporate group are not included in the scope.

Likewise, the analysis is limited to the digital environment and the data ecosystem provided by the company, without considering external information or competitive aspects outside the company. The direct implementation of the proposals with suppliers is not considered either; the objective is to generate internal analytical tools, such as dynamic dashboards and segmentations, that support commercial decision-making and the communication of results.

Despite the limitations described below, the project seeks to deliver a ticket clustering and an interactive dashboard that allow the company to better understand its customers' behavior, detect opportunities by segment and support the design of future campaigns.

LIMITATIONS

The volume of available data was one of the main challenges of the project. Despite applying reduction and sampling strategies, the size and complexity of the databases exceeded the processing capacity of the equipment used, which limited the possibility of running certain more advanced analytical methods or extending the analysis to all the years and stores originally planned. In particular, although there was enough historical information to broaden the time horizon, doing so would have meant working with an excessive volume of data that would have hindered both processing and the interpretation of results, which is why the analysis was restricted to the last three years.

Likewise, the limited computing capacity made it difficult to implement more demanding models, such as those based on deep learning (for example, full autoencoders or clustering models such as DBSCAN or HDBSCAN), and forced us to prioritize techniques that balanced accuracy and operational feasibility. Even after reducing the dataset to manageable subsets, execution times remained high, which limited the depth of some tests.

This breadth of data also brought challenges related to data quality and consistency. Records with inconsistent values in units and amounts were identified, especially in products corresponding to redemptions, returns, containers and deposits, which showed negative or zero values with no clear pattern that would allow a consistent business rule to be defined. The same products appeared with positive values in other records, which increased the uncertainty about their interpretation. After consulting with the company's team, it was decided to exclude these subcategories from the analysis, as they were considered problematic and unreliable.

Another relevant limitation was the presence of customers with random or sporadic behavior, particularly those who made a single purchase during the period analyzed. Although this group is part of the real customer base, its behavior introduced a considerable level of noise into the segmentation models, making it harder to detect stable and representative patterns across the customer base.

It is important to clarify that the company defines a loyalty customer as one who participates in the loyalty program and authorizes the use of their purchase data linked to their national ID number. This means that the concept of "loyalty customer" used throughout the project is not necessarily equivalent to a recurring customer of the chain. This broad definition means that the database includes both truly loyal consumers and occasional shoppers, which increases the heterogeneity of the dataset and adds complexity to the analysis.

In the particular case of the dashboard developed in Power BI, the main limitation was related to the magnitude of the transactional volume. Unlike the modeling process, where it was possible to work with a limited sample of stores and years to facilitate algorithm training, a corporate dashboard must cover the complete behavior of the chain, without being restricted to specific subsets. This meant dealing with a flow of thousands of daily transactions, whose direct incorporation into the tabular model resulted in excessive load times, high resource consumption and a poor user experience. To mitigate this problem, we chose to build a pre-aggregated transactional table, sacrificing purchase-by-purchase and customer-by-customer detail, but retaining the information needed to analyze demand by store, subclass and supplier, which was the focus of the dashboard.

Likewise, the last two years were defined as the time horizon, both because of their relevance to the analytical objectives and because of the greater consistency and quality of recent records. These decisions made it possible to strike an appropriate balance between data representativeness, technical performance and dashboard usability, even when working with a table that, in its final form, exceeded eight million records.

METHODOLOGY

The project was developed using an iterative methodology, working in short cycles that allowed the analysis to be adjusted as the process advanced and new needs or limitations arose. The work was organized through weekly agendas, which defined the tasks and objectives for each period. Each week, a follow-up meeting was held with the academic advisor to review progress, resolve questions and agree on next steps.

In contrast, contact with the company was occasional and unstructured. It took place only when it was necessary to clarify questions about the data or validate specific information. This created a more reactive communication flow and required flexibility to adapt the work to the company's response availability.

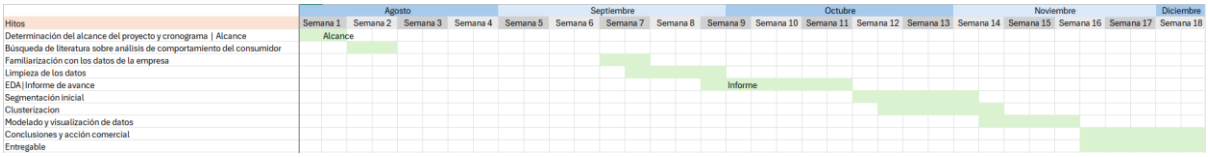
The project was carried out in the following stages:

- Initial definition of the work: the scope was defined and the timeline was drawn up, together with a literature review that established the conceptual framework and guided the selection of tools and techniques.
- Data access and preparation: access to the database was obtained, the data was exported to Jupyter Notebook and an initial cleaning was performed to correct quality, format and consistency issues.
- Exploratory analysis: the behavior, distribution and descriptive metrics of the data were investigated.
- Modeling and segmentation: different methods were applied (RFM, K-Means, DBSCAN, Word2Vec and Autoencoders), evaluated based on prior research and computational limitations.
- Clustering: the most consistent clusters were identified and their characteristics analyzed to interpret the different purchasing behavior patterns.
- Power BI dashboard development: an interactive tool focused on the temporal and aggregate analysis of sales was built, making it possible to visualize seasonality, variations in demand and metrics relevant to decision-making.
- Conclusions and recommendations: conclusions were drawn based on the findings of the analysis, and commercial recommendations were formulated to take advantage of the opportunities detected and guide future strategic decisions.

TIMELINE

The initial project timeline had to be adjusted due to communication difficulties with the company, which caused a delay of approximately four weeks compared to the original plan. Once this issue was overcome, the tasks in the timeline were reorganized, prioritizing data extraction and unification as a fundamental step toward the subsequent stages of cleaning, normalization and exploratory analysis.

Although the delay reduced the time available for technical execution, adding extra work sessions made it possible to recover part of the time and maintain consistency between the defined scope and the stated objectives.



EXPECTED SOLUTION

The proposed solution seeks to provide the company with a robust analytical tool that enables an in-depth understanding of its customers' consumption patterns, identifying differentiated behavioral segments based on the available transactional information.

The project aims to generate a meaningful clustering based on purchase tickets, capable of grouping similar behaviors and revealing different consumption profiles within the supermarket network. This approach would make it possible to detect groups of customers with common characteristics, such as purchase frequency, type of products purchased, average ticket amount and category preferences, providing a clearer view of the dynamics that influence purchasing decisions.

As an expected result, an interpretable segmentation model is to be built, accompanied by an interactive dashboard that facilitates data exploration and the identification of commercial opportunities.

These visualizations will allow the company to:

- Detect recurring purchasing patterns or changes in consumer preferences.
- Identify segments with high growth potential or churn risk.
- Provide actionable information to the marketing and supplier areas, fostering personalized communication, promotion or assortment strategies.

Finally, from an academic perspective, the project aims to demonstrate the feasibility of applying advanced data analysis methodologies to mass retail contexts, contributing to the development of replicable models in other areas of the organization or in future research.

2. OBJECTIVES

General objective

To explore and understand in detail the purchasing patterns observed in the chain based on the available data, in order to generate actionable knowledge that supports commercial and strategic decision-making and facilitates the communication of insights to suppliers.

Specific objectives

- To analyze the temporal dimension of consumption in order to recognize seasonality, trends and relevant variations in demand.
- To explore possible groupings or segmentations of customer tickets based on their purchasing habits, as a descriptive characterization and for analytical purposes.
- To develop interactive visualization tools that facilitate continuous monitoring of customer behavior and the communication of insights, both within the company and to suppliers.
- To provide an executive report with the main findings and strategic recommendations, aimed at strengthening customer relationships and improving commercial management.

One of the initial objectives of the project was to carry out a customer-centered segmentation, with a more individualized approach oriented to each consumer's profile, analyzing their behavior over time. However, this approach proved methodologically unfeasible due to the large amount of data available, which made it impossible to work simultaneously with multiple stores and years of information while maintaining a cross-sectional view of the customer base. In addition, the strong presence of sporadic or low-recurrence consumers (those who made a single purchase or very isolated purchases) introduced a significant level of noise into the models, making it difficult to identify consistent patterns

representative of actual behavior. For this reason, an alternative approach based on purchase tickets was chosen, which made it possible to maintain analytical robustness and obtain a global characterization of consumption without compromising the technical feasibility of the analysis.

3. BUSINESS CONTEXT

The corporate group to which the chain belongs is one of the leading retail groups in the country, with operations in every department and both a physical and digital presence. The company brings together brands in different sectors, such as supermarkets, pharmacies, home goods and appliance stores, as well as online sales channels. Its value proposition focuses on accessibility and proximity, with store formats ranging from large hypermarkets to convenience stores, adapting to the needs of different consumer segments.

One of the pillars of the company's value proposition is its loyalty program, which has a large base of active members. Every transaction made by customers is recorded together with relevant demographic information (age, gender, location, socioeconomic level). In this way, the program not only works as a retention and rewards tool, but also as a strategic source of data that makes it possible to understand purchasing behavior in depth and design more effective commercial actions.

Within the business units, supermarkets are a strategic pillar. The volume of loyalty customers and the richness of transactional information make them the most suitable area for advanced data analysis. Understanding customer behavior in detail is essential to strengthen the company's value proposition, improve decision-making and remain competitive in an increasingly dynamic market.This information also represents an opportunity to strengthen relationships with suppliers, by generating insights that enhance joint planning and the development of shared value propositions.

Although the loyalty program is a valuable source of information for the company, it also has a relevant conceptual and operational limitation: the available data does not strictly represent loyal customers, but only those who registered their national ID number at the time of purchase. According to Oliver (1999), "customer loyalty is the process through which a company develops lasting relationships that increase the likelihood that a consumer will continue buying and maintain a sustained preference for the brand." This definition shows that loyalty implies recurrence and a stable relationship with the company, something that cannot be inferred simply because a customer registers their ID number in an isolated purchase. Real loyalty is reflected in behavior sustained over time; however, the current program collects information only when the customer voluntarily enters their ID number, which does not guarantee that all registered shoppers are actually loyal.

Two main hypotheses explain the lack of consistent purchase registration:

1. Customers who do not enter their ID number, whether due to lack of awareness of the benefits, lack of habit, disinterest or absence of perceived incentives. This group generates anonymous transactions that reduce the traceability of purchasing behavior and weaken the analytical potential of the program.
2. Customers who do enter their ID number but are not truly loyal, that is, occasional shoppers who visit only a few times a year. Although they are registered in the database, they do not show loyal or recurring behavior, which introduces noise into analyses aimed at identifying consistent patterns.

These limitations represent a strategic challenge for the company, which needs to strengthen its loyalty mechanisms, improve data capture and develop more effective incentives to ensure that purchases are correctly linked to each consumer.

4. CLUSTERING MODEL

4. 1 RESEARCH

More than 90% of the customers who shop at the chain are part of the loyalty program, which represents an extremely valuable source of information for the company. This high level of enrollment makes it possible to record consumers' purchasing behavior in detail, creating a great opportunity to analyze patterns, understand habits and extract strategic insights that contribute to more informed, data-driven decision-making.

Segmentation and market strategy

"Studies have shown that it is more profitable for a company to retain a customer than to try to win a new one. The value of retaining a customer is mainly due to the fact that it has lower costs than those involved in winning a new customer. Therefore, caring for relationships with current customers is a better investment. Economically, it is 6 to 7 times more expensive to acquire a new customer than to retain an existing one. And by increasing investment in customer retention by just 5%, profit increases of 5 to 95% could be achieved. For this reason, companies are forced to invest time, money and effort in learning as much as possible about their customers" (Anita Rojas Huamán, 2020).

This makes even more sense considering that markets can reach saturation levels, where acquiring new customers becomes increasingly difficult and costly. In contrast, existing customers are constantly evolving, changing their needs, preferences and purchasing behavior over time. Being able to anticipate and predict these changes becomes a key factor in keeping the company relevant, strengthening its relationship with customers and generating a sustainable competitive advantage in the market.

To achieve this goal, customer segmentation becomes a fundamental tool. Through this process, it is possible to gain an in-depth understanding of consumers, identifying groups with similar characteristics, behaviors and needs. This understanding makes it possible not only to design more effective and personalized commercial strategies, but also to predict relevant information about their future behavior, anticipating trends, preferences and potential business opportunities. "It can be defined as 'the practice of dividing a customer base into groups of individuals who are similar in specific ways relevant to marketing, such as age, gender, interests, spending habits, etc.'" (Anita Rojas Huamán, 2020).

In addition to improving retention, customer segmentation makes it possible to understand the value each consumer brings to the organization and to allocate resources more strategically and efficiently. By identifying behavior patterns, spending levels and purchase frequency, companies can distinguish between high-value and low-value customers, prioritizing commercial, communication and loyalty efforts according to their impact on profitability.

"From the company's point of view, in order to continue making a profit, they need to direct their attention to the customers who consume their products or services most frequently or in large volumes, in order to create fruitful groups. In other words, customers have different levels of importance, and in order to remain in the market, companies must distribute their attention unequally, which means they must shift their attention to the customers who generate the greatest profits for the company." (Anita Rojas Huamán, 2020)

Benefits of customer segmentation:

- Precisely defining buyer personas, or representative profiles of ideal customers, makes it possible to identify how they search for, buy and use a product or service. This knowledge enables better prioritization and targeting of commercial actions, the design of more effective marketing messages and improved customer retention and loyalty. (Contentsquare, 2024)
- Brand differentiation is achieved through segmentation, which makes it possible to align the value proposition with the specific needs of each target audience, thus strengthening the company's identity and positioning in the market (Adobe Experience Cloud Team, 2025).
- It enables sustainable growth "when strategies are adapted to the unique needs of each audience" (Adobe Experience Cloud Team, 2025).
- "It maximizes return on investment by directing efforts where they are most effective" (Adobe Experience Cloud Team, 2025).
- As mentioned above, it helps increase customer retention, understand the customer base in greater depth and optimize resource allocation.

In a context of growing competition and market saturation, business survival is becoming increasingly challenging. In this scenario, customer segmentation has become an essential tool for any business, as it makes it possible to understand consumers' needs and behaviors more precisely, optimize commercial strategies and strengthen relationships with the most valuable segments.

RFM

Among the many models used for customer segmentation, the RFM (Recency, Frequency, Monetary) model stands out for its simplicity and effectiveness. This model evaluates customers based on how recent their last purchase was (Recency), how often they buy (Frequency) and how much they spend (Monetary). Each of these dimensions provides valuable information about a customer's level of engagement and the value they represent for the company. When combined, these metrics make it possible to build a robust profile that can be used to segment customers and guide more precise and effective marketing strategies.
(Yohanni Syahra, Abdul Fadlil, Herman Yuliansyah, 2025)

The variables are calculated as follows:

- Recency = Date of the last transaction in the dataset − Date of the customer's last purchase. It represents the number of days elapsed since the customer's last purchase.
- Frequency = Total number of transactions made by the customer during the analysis period
- Monetary = ∑ (Total amount of each of the customer's transactions)

"The analysis consists of classifying customers by their value based on the three variables. To do so, scales are built based on these variables, giving each customer a value according to the percentile they fall into. The most common approach is to scale by quintiles. That is, all variables are assigned a value between 1 and 5, with 1 being the worst score and 5 the best" (Anita Rojas Huamán, 2020).

In this case, the most profitable and ideal customers within the RFM model are those with the highest scores in all three dimensions, that is, a 5-5-5 value. These customers buy frequently, have purchased recently and spend high amounts, which makes them the segment of greatest strategic value for the company.

"For example, a customer in the top 20% of most recent purchasers, in the 2nd 20% by purchase frequency and in the 4th 20% by total purchase value would be assigned segment 542, (R)5 (F)4 (M)2. We can see it in the chart:" (Guillermo Córdoba, 2011)

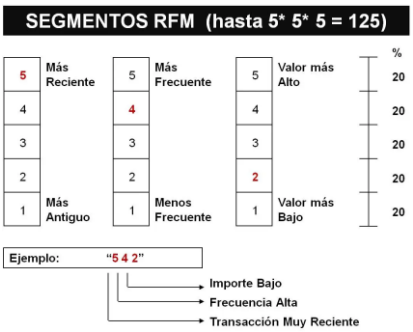


Figure 4.1.1 - Example of RFM segmentation

Clustering

Clustering is a widely used technique in unsupervised machine learning, whose purpose is to identify groups of elements with similar characteristics within a dataset, in order to obtain new insights and facilitate the generation of predictions. This approach has multiple applications in various fields, including image recognition, customer segmentation and anomaly detection. Among the most widely used algorithms for this type of task are DBSCAN and K-Means (Taylor, 2024).

Clustering: K-Means

The K-Means algorithm belongs to the family of partition-based clustering methods. It assigns each point in the dataset to a single cluster, with no overlap between clusters and no unassigned points; as a result, it generates an exhaustive and mutually exclusive partition of the dataset. Due to its simplicity, interpretability and computational efficiency, K-Means is widely used in contexts with large volumes of information (Idrees, 2024).

The algorithm works in the following steps:

1. Choose K: define the number of clusters (K), a hyperparameter set by whoever applies the algorithm.
2. Initialize centroids: randomly place K centroids (one per cluster) in the data space.
3. Assign points to clusters: each point is assigned to the nearest centroid according to Euclidean distance.
4. Update centroids: recalculate the centroids by taking the mean of all the points assigned to each cluster.
5. Repeat: repeat the process until the centroids no longer change their position significantly (Idrees, 2024).

The mathematical goal of K-Means is to minimize the objective function:

$$J = \sum_{i=1}^{K} \sum_{x \in C_i} \|x - \mu_i\|^2$$

where K is the number of clusters, C_i is the set of points belonging to cluster i, μ_i is the corresponding centroid, and ‖x−μi‖^2 represents the squared distance between point x and its centroid (Idrees, 2024).

The main advantages of K-Means include its simplicity, computational efficiency, ability to scale to large volumes of data and ease of interpretation, since its results can be easily visualized (Idrees, 2024).

However, the algorithm has some limitations: it requires the number of clusters to be defined in advance, it is sensitive to the presence of outliers (which can significantly alter the location of the centroids), and it assumes that clusters are spherical and of similar size, which hinders its performance when the data has more complex structures (Idrees, 2024).

K-means: elbow method

The most common technique for identifying the optimal number, k, of clusters into which to divide a dataset with K-means is the elbow method. This method seeks to identify the point at which the decrease in within-cluster variance slows down, so that increasing the number of clusters would not significantly improve the compactness of the data (Rodriguez, 2023).

To obtain the optimal k, the k-means algorithm is run with different numbers of clusters and, for each k, the inertia or WCSS (within-cluster sum of squares) is calculated, which corresponds to the sum of the squared distances between the points in a cluster and its centroid.

When plotting the inertia values as a function of k, the value where the elbow forms is chosen, such that "increasing the value of k does not lead to a significant reduction in WCSS" (Trovar, 2023).

K-means: silhouette score

The silhouette score is also a measure used to evaluate how well the clusters are formed after applying unsupervised learning algorithms such as k-means and, based on that, determine the optimal number of clusters. For each data point, it measures how well placed it is within its own cluster, by calculating the average distance from the point to all points in the same cluster, and how separated it is from the other clusters, by calculating, for each cluster other than its own, the average distance from the point to that cluster (Geeks For Geeks, 2025).

$$a(i) = \frac{1}{|C(i)| - 1} \sum_{\substack{j \in C(i) \\ j \neq i}} d(i,j) \qquad b(i) = \min_{j \neq i} \left(\frac{1}{|C(j)|} \sum_{j \in C(j)} d(i,j) \right)$$

With this, it builds a number between –1 and 1: a value close to 1 means that the point is very well assigned to a cluster and that the clusters are distinguishable, a value close to 0 means it lies on the border between two clusters and the clusters overlap, and a negative value indicates that the point is probably misassigned (Trovar, 2023).

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

To obtain the measure for the algorithm as a whole, and thus identify the optimal number of clusters, the average score across all points is computed and compared for the different values of k (Geeks For Geeks, 2025).

Clustering: RFM + K-Means

Combining the RFM model with the K-Means algorithm is one of the most widely used strategies for customer segmentation, given its ability to uncover natural patterns within large volumes of data. While the RFM model quantifies consumer behavior based on recency, frequency and monetary value, K-Means groups customers into homogeneous segments based on the similarity of those behaviors. This integration makes it easier to identify groups that are not always evident through traditional analysis. Several studies have demonstrated the effectiveness of this methodology in different sectors: Ma (2022) applied K-Means to RFM data in e-commerce, improving segmentation and targeting strategies, while Paul and Ramanan (2019) showed an increase in customer retention in the hotel industry by combining both techniques. Similarly, more recent research (Tabianan et al., 2022; Nasyuha et al., 2022) highlights that using RFM and K-Means together allows retail companies to identify high-value

customers, recognize inactive consumers and adapt their marketing strategies based on actual behavior. In short, this integration not only enriches the decision-making process, but also promotes sustainable growth through personalized, data-driven strategies. (Yohanni Syahra, Abdul Fadlil, Herman Yuliansyah, 2025)

Clustering: DBSCAN

DBSCAN (Density-Based Spatial Clustering of Applications with Noise), on the other hand, is a density-based algorithm that groups points that are sufficiently close to each other and classifies as noise those that are isolated or located in low-density regions (Idrees, 2024).

The DBSCAN procedure can be summarized as follows:

1. Define parameters: set the maximum distance (ε) and the minimum number of points (MinPts) required to form a dense region.
2. Identify core points: determine the points that have at least MinPts neighbors within radius ε.
3. Form clusters: connect the core points with their neighbors to form the clusters.
4. Classify noise: any point that does not belong to any cluster is classified as noise (Idrees, 2024).

Formally, a point p is considered a core point if there is a set of points N(p) such that:

$$N(p) = \{q \in D \mid dist(p,q) \leq \varepsilon\}, \quad \text{y} \quad |N(p)| \geq MinPts$$

where D is the dataset, dist(p,q) is the distance between points p and q, and ε is the maximum radius considered for belonging to the cluster (Idrees, 2024).

Unlike K-Means, DBSCAN does not require the number of clusters to be specified in advance, but determines it automatically based on the density of the data. In addition, it identifies outliers as noise, which makes it more robust to atypical observations. It is also able to detect clusters of arbitrary shapes, not necessarily spherical, adapting better to complex distributions (Idrees, 2024).

However, its performance depends heavily on the values chosen for the hyperparameters ε and MinPts; an inappropriate choice can lead to unrepresentative results. Likewise, its effectiveness decreases when densities vary significantly between clusters or when the number of dimensions increases, due to the loss of relevance of the concept of distance in high-dimensional spaces (Idrees, 2024).

In summary, K-Means is recommended when the approximate number of clusters is known, the data is well separated or the dataset is large. In contrast, DBSCAN is more suitable when the number of clusters is unknown, the structures are irregular or the dataset contains outliers that could distort the results of K-Means (Idrees, 2024).

Autoencoder

Autoencoders are unsupervised learning algorithms designed to learn useful representations of data with little or no supervision. According to Tschannen, Bachem and Lucic (2018), learning meaningful representations with little supervision is one of the central challenges of artificial intelligence. These neural networks learn a mapping from high-dimensional observations to a smaller representation space, from which the original observations can be approximately reconstructed.

The main goal of an Autoencoder is to transform the input into an output with as little distortion as possible (Zhang, 2018). For this reason, they are widely used for data compression. Since they reduce dimensionality in a non-linear way, they make it possible to discover more compact and relevant representations. This ability arises from imposing structural constraints, such as limiting the number of

hidden units, which forces the model to identify the most important regularities in the dataset (Ng, 2011).

The use of Autoencoders can also improve segmentation quality, by making it easier for clustering algorithms to identify less evident patterns. In particular, Variational Autoencoders (VAEs) promote the *disentanglement* (*separation into independent factors*) of the latent variables, generating abstract and concise representations of the data that are useful for various downstream tasks (Tschannen et al., 2018). In addition, these models can incorporate mixture structures in the latent variables, so that each component corresponds to a specific category within the dataset (Tschannen et al., 2018).

Word2Vec

On the other hand, *Transfer Learning*, through pre-trained models such as Word2Vec (W2V), makes it possible to leverage advanced vector representations of words (Goldberg & Levy, 2014). Word2Vec, developed by Mikolov et al. (2013), represents words in a continuous vector space where words that share similar contexts are located close to each other. These representations have been shown to capture semantic meaning and to be useful in multiple natural language processing tasks (Rong, 2014). In addition, Word2Vec is one of the fundamental precursors of modern Natural Language Processing architectures and has been integrated as a component in later, larger-scale models, including Large Language Models such as GPT (Radford et al., 2018).

This methodology is especially valuable in business contexts, where processing large volumes of data involves high time and computational costs. In this sense, Word2Vec is an effective strategy for reducing feature dimensionality, both by decreasing computation time and by improving the performance of classification tasks (Ma & Zhang, 2015). By grouping similar words, the original dimension is projected into a new, smaller space, which speeds up the analysis and reduces processing costs.

In summary, both Autoencoders and Word2Vec seek to generate more compact and meaningful representations of data, although they differ in purpose and application. While Autoencoders are used to discover underlying structures through non-linear reconstructions, Word2Vec is specifically designed for language, generating semantic representations from context.

A key aspect of Word2Vec is that the relationships learned between vectors can be evaluated through measures of spatial proximity, among which cosine similarity is the most widely used in the literature. This metric makes it possible to quantify the degree to which two elements share context patterns, regardless of the magnitude of their vectors, focusing only on their orientation in the vector space. As Turney and Pantel (2010) point out, this approach is especially suitable for distributed representations, since it captures semantic affinities emerging from co-occurrences without requiring prior categorical information. Consequently, cosine similarity is a standard tool for identifying semantic proximity in embedding-based models, and its application in this work made it possible to evaluate the consistency of the relationships between products derived from the model's training process (Mikolov et al., 2013; Rong, 2014).

Word2Vec + K-Means

The relationship between Word2Vec and K-means can be seen in the joint use of both methodologies to optimize the handling of large volumes of text. Word2Vec makes it possible to obtain word vectors and measure their similarity, while K-means groups these vectors based on their proximity, reducing

the dimension of the original space (Ma & Zhang, 2015). In this way, K-means acts as a dimensionality reduction strategy by condensing the vector space into semantically coherent categories.

Autoencoder + K-Means

Similarly, Variational Autoencoders (VAEs) share with K-means the search for latent structures that reflect natural groupings. For Tschannen, Bachem and Lucic (2018), the clustering structure is a useful *meta-prior* for obtaining more informative representations, and it can be encouraged by incorporating mixture distributions that promote the formation of clusters.

Autoencoder + DBSCAN

Chen, Zhang and Liu (2021) propose combining a VAE with the DBSCAN algorithm within an *active learning*approach. The authors highlight that the VAE adequately represents high-dimensional data in a reduced latent space, while DBSCAN is effective at identifying noise and discovering clusters without the need to specify the number of classes in advance. Combining both methods improves efficiency and accuracy in sample selection in high-dimensional environments.

Word2Vec + DBSCAN

Similar to the combined VAE and DBSCAN approach, it is also common to use word embeddings, for example those generated by Word2Vec, and then apply DBSCAN to the word vectors to discover semantic groupings without having to set the number of clusters in advance. In this regard, the study "Influence of various text embeddings on clustering performance in NLP" notes that DBSCAN outperforms K-Means when applied to Word2Vec embeddings (Saha, 2023).

This combination makes it possible to group words or terms that have similar contexts in the vector space, facilitating the identification of relevant semantic clusters without requiring the number of groups to be assigned beforehand.

Calinski-Harabasz index

The Calinski-Harabasz index, also known as the Variance Ratio Criterion, can be used to evaluate models when the true labels of the data are unknown. The index is the ratio of the sum of between-cluster dispersion to within-cluster dispersion for all clusters (where dispersion is defined as the sum of squared distances). (Scikit-learn Developers, n.d.)

For a dataset E of size n_e that has been divided into k clusters, the Calinski-Harabasz score s is defined as the ratio of the mean between-cluster dispersion to the within-cluster dispersion:

$$s = \frac{\operatorname{tr}(B_k)}{\operatorname{tr}(W_k)} \times \frac{n_E - k}{k - 1}$$

where $\operatorname{tr}(B_k)$ is the trace of the between-group dispersion matrix and $\operatorname{tr}(W_k)$ is the trace of the within-cluster dispersion matrix, defined as:

$$W_k = \sum_{q=1}^{k} \sum_{x \in C_q} (x - c_q)(x - c_q)^T$$

$$B_k = \sum_{q=1}^{k} n_q (c_q - c_E)(c_q - c_E)^T$$

where:

- C_q is the set of points in cluster q,
- c_q is the center of cluster q,

- c_E is the center of the full set E,
- n_q is the number of points in cluster q.

(Scikit-learn Developers, n.d.)

Calinski-Harabasz index scores are higher when clusters are dense and well separated, which matches the standard concept of what a cluster should be. In addition, it is fast to compute. However, it has an important drawback: the index value tends to be higher for convex clusters compared to other types of clustering, such as density-based methods like those obtained with DBSCAN. (Scikit-learn Developers, n.d.). In line with this characteristic, the index tends to perform especially well when applied together with clustering methods that assume spherical partitions in the variable space, such as K-means, for which its mathematical structure is particularly well suited.

PCA

PCA (principal component analysis) is a technique for reducing the dimensionality of datasets by reducing them to their principal components, so that most of the information is preserved. It consists of transforming potentially correlated variables into a smaller set of variables (IBM, 2025).

The technique uses matrix operations and transforms the original dataset into a new one containing only the principal components: the eigenvectors indicate the direction of variance, while the eigenvalues act as coefficients of the eigenvectors and denote the importance of these directions. In this way, the algorithm takes as the first principal component the eigenvector with the largest eigenvalue, since it maximizes the variance in that direction. The second principal component is obtained in a similar way, but must be orthogonal to the first in order to maximize the explained variance. The procedure is repeated for the remaining principal components and the dimensions to be considered (IBM, 2025).

t-SNE

t-SNE (t-distributed Stochastic Neighbor Embedding) is a non-linear dimensionality reduction technique for exploring and visualizing high-dimensional data (Awan, 2024).

It differs from PCA in that it is non-linear and preserves the local (pairwise) similarities between data points in a lower-dimensional space, that is, the relationships between data points. In contrast, PCA aims to preserve variance in a lower-dimensional space (Awan, 2024).

The algorithm consists of three steps. First, it computes similarities in the high-dimensional space: for each pair of points, it calculates how likely they are to be neighbors using a Gaussian distribution. It then tries to reproduce these similarities in the low-dimensional space and computes the neighborhood probabilities in that reduced space. Finally, it minimizes the difference between both distributions using gradient descent, so that the similarities between pairs of points are as close as possible to those in the original space (Awan, 2024).

ADF

The Augmented Dickey-Fuller (ADF) test is a statistical test used to check whether a time series is stationary or whether, on the contrary, it follows a "long memory" pattern in which shocks tend to persist. In simple terms, the ADF tests the hypothesis that the series has a unit root, that is, that it behaves like a random walk and does not return to a stable average level, against the alternative that it is stationary and mean-reverting. In practice, this also makes it possible to assess whether the apparent trend of the series is truly deterministic (for example, a linear trend that can be modeled explicitly) or whether it is due to a stochastic component linked to the unit root. To do so, the test fits a model that relates the current change in the series to its own past, allowing additional lags to be included to capture the temporal dynamics, and uses a special statistic to decide whether or not to

reject the presence of a unit root (Dickey and Fuller, 1979; Dickey and Fuller, 1981). In empirical applications, the ADF has become a standard tool for analyzing the stationarity of economic and financial series before specifying more complex time series models (Mushtaq, 2011).

t-test

The one-sample t-test is a basic statistical test used to check whether the observed mean of a variable differs significantly from a reference value, typically zero. Applied to the seasonal component of a time series, the test contrasts the null hypothesis that the mean of that component is zero (that is, that there is no systematic seasonal effect) against the alternative that the mean is different from zero. Intuitively, if the test yields a small p-value, it is concluded that the seasonal component tends to deviate from zero consistently, which suggests the presence of seasonality that cannot be attributed to chance. This idea is based on the standard use of the one-sample t-test to test population means described in classic applied statistics texts, such as Montgomery and Runger (2014).

ANOVA

One-way analysis of variance (ANOVA) is a technique that makes it possible to compare the means of several groups simultaneously, evaluating whether the differences observed between them can be explained by chance alone or whether they are statistically significant. In the context of a daily sales series, ANOVA can be used, for example, to test whether mean sales are the same for all days of the week or whether at least one day behaves differently. Under the null hypothesis, all means are equal (the day of the week does not affect sales); a small p-value indicates that this hypothesis is rejected and that the weekly pattern is relevant, which constitutes evidence of intra-week seasonality. This interpretation follows the classic treatment of one-way ANOVA in experimental design textbooks, such as Montgomery (2017).

Sparsity index

In the analysis of high-dimensional data, especially in transactional contexts, it is common to evaluate the sparsity index. This indicator measures the proportion of null or zero values within a matrix or dataset, reflecting how sparse the available information is. A highly *sparse* matrix has many zeros and few meaningful values, which affects the ability of algorithms to identify latent structures, similarity patterns or well-defined clusters. In the field of machine learning, sparsity is particularly relevant in representations such as customer-product or document-word matrices, where most interactions do not occur. The literature suggests that distributed representation techniques, such as Word2Vec, can help mitigate some of the effects of sparsity by learning dense embeddings that capture semantic relationships even in sparse contexts (Mikolov et al., 2013; Rong, 2014), although their effectiveness depends heavily on the quantity and quality of the available data.

4. 2 EXPLORATORY DATA ANALYSIS

• 4. 2. 1 DATA EXTRACTION

For the development of the project, the company granted direct permissions and access to its Snowflake infrastructure, assigning each team member an individual username and password. In this way, the work was carried out on real-time data, covering information from 2015 to the most recent available date, with daily updates from all business units.

The company enabled only the specific tables needed for the analysis, so no initial filtering at the table-selection level was necessary. The main task was to determine which columns to use, which involved a

process of understanding the data based on its meaning and on the structure of the Entity-Relationship Model (Appendix 1) provided.

Selecting the final dataset was a considerable challenge, since the tables were very large and the team's initial intention was to make use of all the available information without any cuts. However, after a joint analysis with the company and in coordination with the advisor, it was determined that it was necessary to reduce the volume of data to make the work feasible.

Consequently, it was decided to limit the sample to the last two complete years (2023 and 2024) and three specific stores, strategically selected because they represent three different business profiles: a hypermarket, a supermarket and a convenience store (proximity format). Even with this reduction, the volume was still high, so an additional temporal segmentation criterion was adopted:

Year 2023 → used for the exploratory data analysis (EDA).

Year 2024 → used for comparison purposes, to evaluate the consistency and stability of the behavior patterns observed in 2023 and determine whether the identified trends held over time.

This approach made it possible to work with representative and up-to-date information, maintaining the diversity of store profiles, while ensuring the technical feasibility of the analysis within the available resources.

Finally, the consolidated dataset consisted of a set of columns that make it possible to integrate the information consistently:

- TICKET: ticket ID
- TIEM_DIA_ID: date ID
- HORA_HORA_ID: hour-of-day ID
- GOEG_LOCL_COD: code of the store where the transaction took place
- GOEG_LOCL_DESC: description (name) of the store where the transaction took place
- ARTC_ARTC_ID: item ID
- CLASE: ID of the class to which the item belongs
- CLASS_NAME: name of the class to which the item belongs
- SUBCLASE: ID of the subclass to which the item belongs
- SUB_NAME: name of the subclass to which the item belongs
- FOPA_FOPA_ID: payment method ID
- FOPA_FOPA_DESC: payment method description
- VNTA_IMPORTE_CON_IVA: item amount (VAT included)
- VNTA_UNIDADES: number of units of the item
- VNTA_FLAG_PROMO: binary field indicating whether the item was purchased on promotion (0 = no promo; 1 = promo)
- SOCI_SOCI_ID: customer ID
- CLIE_CLIE_SEXO: binary field indicating the customer's sex (F = female; M = male)
- SCENTIA_EDAD: customer's age at the time of the transaction
- SCENTIA_NSE: customer's socioeconomic level
- ARTC_OFN1_ID: binary field for type 1 product offers
- ARTC_OFN2_ID: binary field for type 2 product offers
- ARTC_OFN3_ID: binary field for type 3 product offers
- ARTC_COMB_ID: ID of product-combination offers

-	TICKET	TIEM_DIA_ID	HORA_HORA_ID	GEOG_LOCL_COD	GEOG_LOCL_DESC	ARTC_ARTC_ID	CLASE	CLASS_NAME	SUBCLASE	SUB_NAME	FOPA_FOPA_ID	FOPA_FOPA_DESC
Count	6,707,608	6,707,608	6,707,608	6,707,608	6,707,608	6,707,608	6,707,608	6,707,608	6,707,608	6,707,608	6,707,608	6,707,608
Unique	1,306,196	363	NaN	NaN	3	NaN	NaN	380	NaN	1,366	NaN	63
Top	202312112586659839	2023-12-23	NaN	NaN	Hiper Tres Cruces	NaN	NaN	GASEOSAS	NaN	BOLSAS DE MERCADOS	NaN	EFECTIVO
Freq	170	26,747	NaN	NaN		NaN	NaN	317,391	NaN	242,973	NaN	1,689,111
Mean	NaN	NaN	18	166	5,165,180	134,400	396,329	NaN	3,963,287,000	NaN	44	NaN
Std	NaN	NaN	4	95	NaN	97,375	346,774	NaN	3,467,739,000	NaN	29	NaN
Min	NaN	NaN	2	115	NaN	1	-1	NaN	-1	NaN	1	NaN
25%	NaN	NaN	15	115	NaN	23,436	160,009	NaN	1,600,090,000	NaN	31	NaN
50%	NaN	NaN	18	115	NaN	112,376	280,001	NaN	2,800,010,000	NaN	32	NaN
75%	NaN	NaN	21	115	NaN	226,072	710,001	NaN	7,100,010,000	NaN	45	NaN
Max	NaN	NaN	25	419	NaN	440,185	2,010,001	NaN	2,010,001,000	NaN	169	NaN



For the dashboard implementation, a direct connection to the company's database was established, which makes it possible to feed the visualization with real, up-to-date information automatically, ensuring that the indicators reflect sales and transaction behavior in real time.

• 4. 2. 2 DATA CLEANING

Cleaning consisted of identifying missing data and removing it, since it represented a small percentage and did not affect the representativeness of the dataset; discarding columns considered irrelevant to the objectives of the analysis; dealing with outliers and erroneous data, removing records whose amount exceeded realistic values for a purchase in the context analyzed; and merging rows in cases where the same item was recorded on separate lines within the same transaction. To do this, those lines were consolidated by adding up the quantities and recording them in a single row.

The missing data analysis showed that its proportion was low (less than 10%) and that it was evenly distributed across the three stores, so it did not create biases between the types of stores analyzed. Since the incidence was low (below 10% in all cases) and did not affect the representativeness of the dataset, we chose to remove incomplete records rather than impute values, thus ensuring that each transaction kept complete demographic information linked to an identified customer. The columns with missing values corresponded to key demographic variables: sex, age and socioeconomic level (SEL). By removing only these isolated records, the integrity of the dataset was maintained and it was ensured that each sale could be assigned to a customer profile, a necessary condition for later carrying out a coherent segmentation of purchasing behavior.

The "VNTA_FLAG_PROMO" column was removed, since 100% of the records had a value of zero, indicating that this variable contained no useful or differentiating information about sales.

Sales amounts included negative values and amounts equal to zero, which do not represent real transactions. Since these cases were few and probably corresponded to recording errors or occasional returns, it was decided to remove them from the dataset to avoid distortions in the study of purchasing behavior.

The tickets had a structural inconsistency, where the same purchase appeared fragmented across multiple lines. In many entries, instead of recording a single line indicating the total quantity purchased of a product, the system generated several identical rows, each representing one unit of the same item. To ensure record consistency and optimize the database, these duplicate lines were merged, consolidating those that matched the same product, ticket and time. In these cases, the corresponding quantities were added up and kept in a single row representing the total units purchased. This situation may be due to operational differences in the checkout recording process, since some cashiers scan repeated products one by one, while others choose to scan the item once and manually enter the total number of units. This variability in the procedure would explain the duplication observed in certain tickets.

Item amounts showed outliers in their distribution. Based on the quartile values (Q1 = 49 and Q3 = 149), the upper outliers in the amounts were examined in order to detect possible anomalies. Several records with excessively high amounts were identified that, without a doubt, represented data entry errors or system inconsistencies (for example, the sale of 45 main dishes for an amount of $99,999,999). It was determined that amounts above $4,000 corresponded to very particular cases, either due to recording errors or to atypical transactions that did not reflect customers' usual purchasing behavior. For this reason, we chose to remove from the dataset the item records whose amount exceeded that value, thus ensuring the consistency and realism of the data analyzed. These records represented 0.7% of total transactions.

In addition, the few cases in which the sex variable did not have the values "male" or "female" were removed, as well as records in which the customer's age was under 18 or over 100, as they were considered inconsistencies or values not representative of the study population.

- ## 4. 2. 3 EXPLORATORY ANALYSIS

(Consider the exploratory analysis performed and its details)

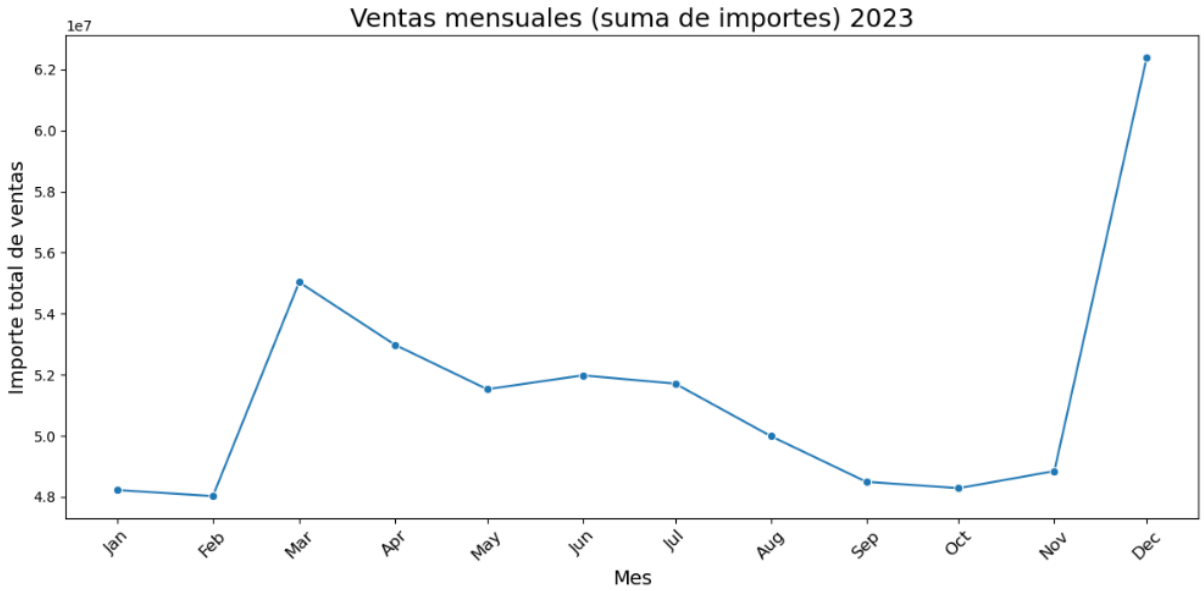


Figure 4.2.3.1 - Sum of sales amounts by month, 2023

The monthly evolution of sales shows that, during the first months of the year, values remain stable at relatively low levels, which can be explained by the fact that at that time many customers travel to the east coast or other summer destinations and therefore reduce their consumption at supermarkets in the capital and in their usual neighborhoods. In March there is a significant rebound, probably linked to the return of daily activity and the start of the school year.

From April onward there is a downward trend that lasts until September, reaching one of the lowest levels in October. This decline can be explained by seasonal and economic factors: these are months without holidays or events that drive consumption, household purchasing power is tighter in the absence of extra income, and in spring it is common for part of the spending to go to other categories such as clothing, leisure or consumption outside the home.

Finally, December marks a turning point with abrupt growth and the highest peak of the year, reflecting the impact of the end-of-year holidays, when supermarket consumption multiplies. This behavior confirms that the business is highly dependent on specific moments in the calendar and highlights the importance of designing commercial and promotional strategies that sustain activity during the "low" months before the high season.

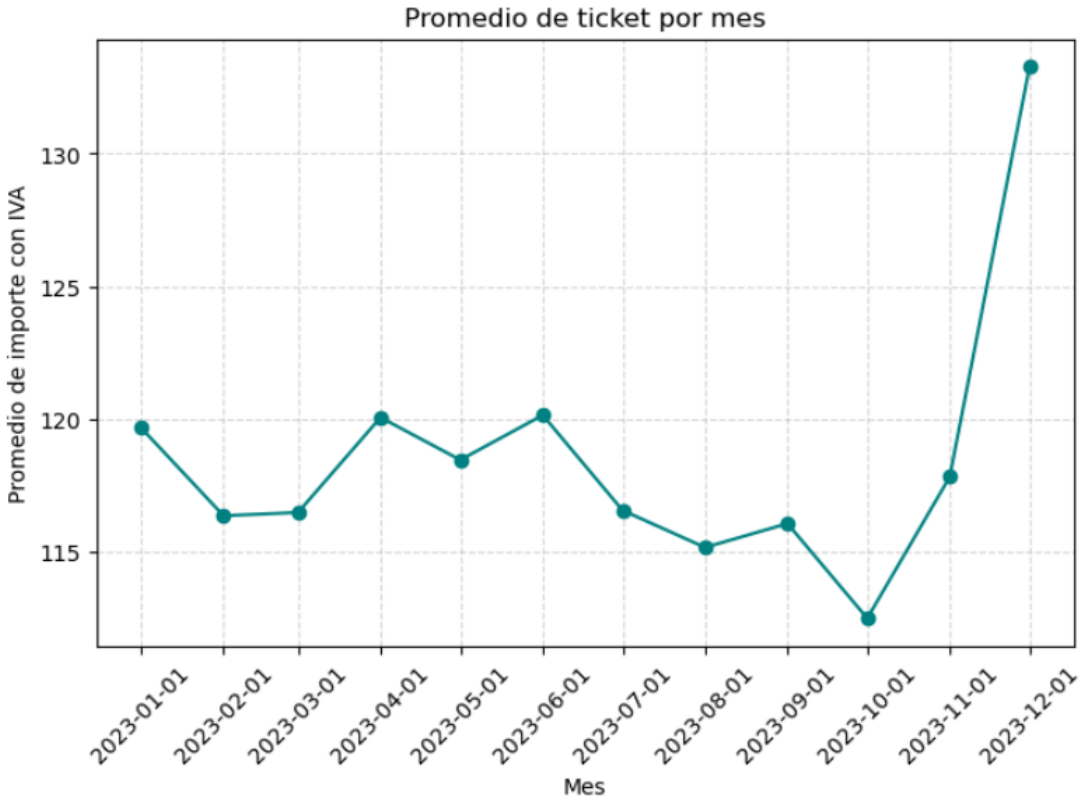


Figure 4.2.3.2 - Average ticket amount by month, 2023

This chart complements the analysis of monthly sales and shows that the variations observed throughout the year are not only due to changes in the number of tickets issued, but also to changes in the average purchase amount per ticket. This shows that sales results combine both the level of customer traffic and individual spending on each visit.

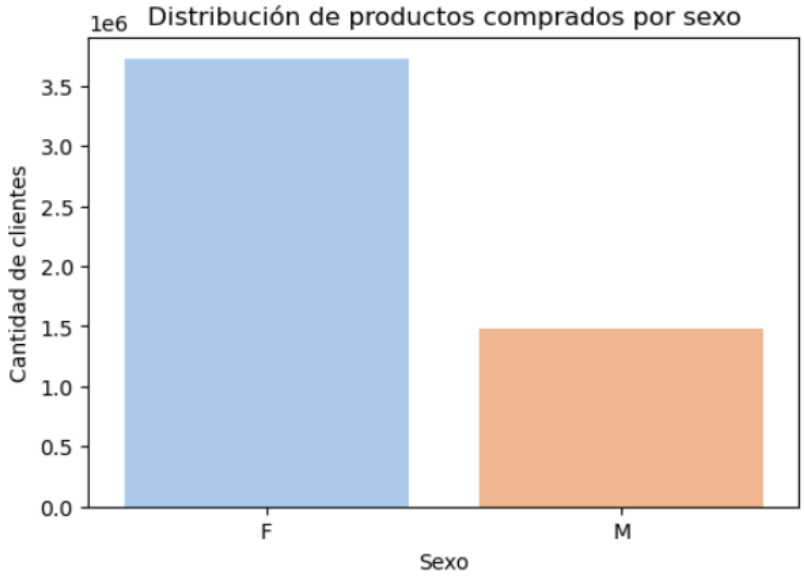


Figure 4.2.3.3 - Distribution of products purchased by sex, 2023

The difference observed in the distribution of products purchased by sex could be due to several factors, which at this stage can only be put forward as hypotheses. A first explanation may be linked to

shopping roles within households, where it is common for women to take on the main responsibility for supermarket shopping, which would increase the number of transactions associated with them.

Another possible explanation is related to registration in loyalty programs. The membership card is often issued in the name of a single person in the household, which concentrates all the family's purchases in that person's profile. Since in many households the responsibility for shopping usually falls on the woman, it is logical that she would appear as the holder of the loyalty registration.

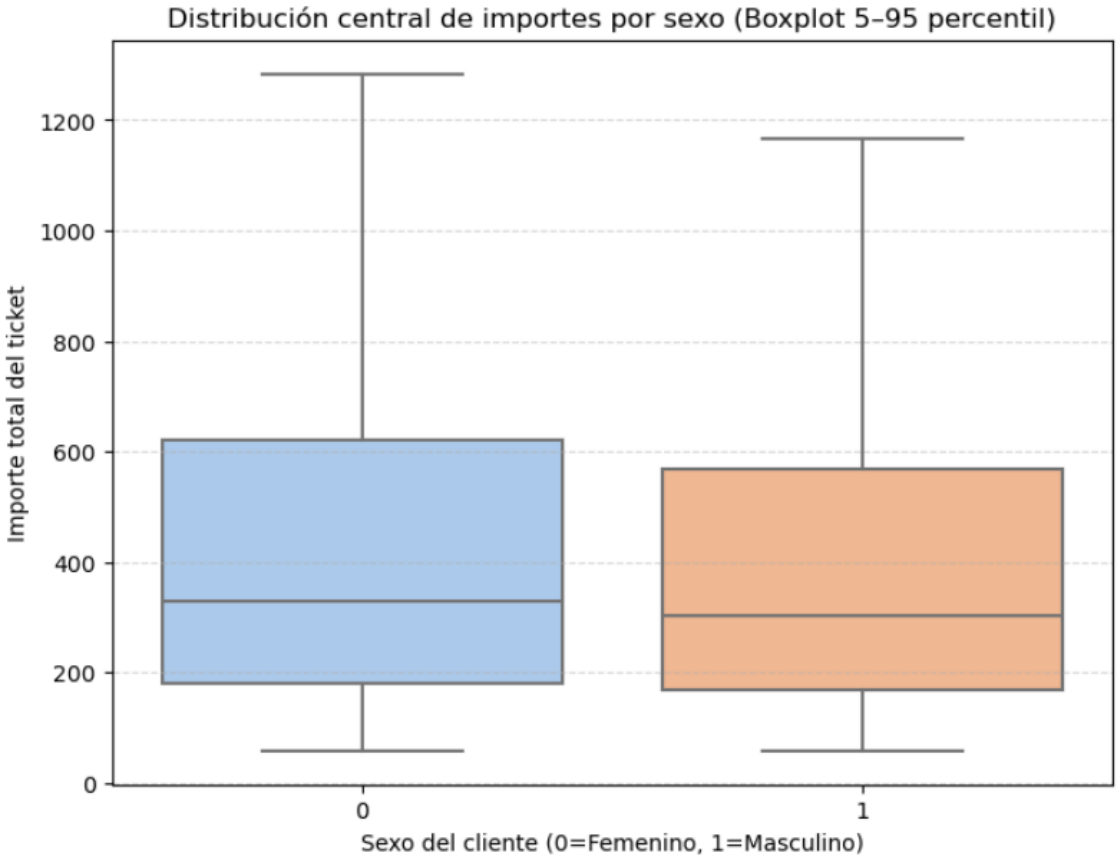


Figure 4.2.3.4 - Boxplots between the 5th and 95th percentiles comparing men and women, showing the central distribution of amounts per ticket

These boxplots show that, in terms of purchase amounts, the behavior of men and women is very similar.

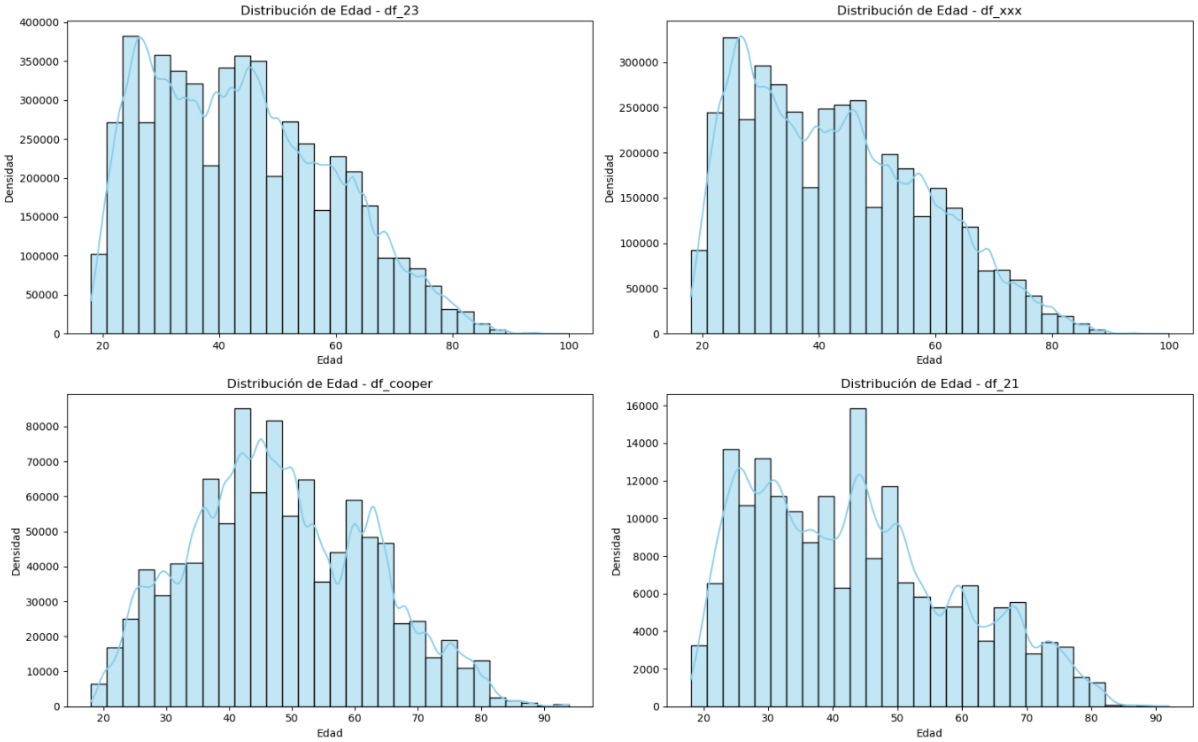


Figure 4.2.3.5 - Sales distributions by age, for 2023, across the three stores studied and for each store individually

- Df_23: transactions from the 3 supermarkets combined.
- Df_21: transactions from the 21 de Septiembre convenience store (Punta Carretas).
- Df_cooper: transactions from the Cooper supermarket (Carrasco Norte).
- Df_xxx: transactions from the Tres Cruces hypermarket.

The analysis of the age distribution makes it possible to identify clear differences between the different supermarket formats. In the case of the hypermarket, there is a greater concentration of customers in younger and middle-aged ranges, mainly between 25 and 45 years old. This suggests that this format is more attractive to active families who make larger, more planned purchases. Another possible reason for this age distribution may be that the Tres Cruces shopping mall, where the hypermarket is located, also serves as the main bus terminal in the country's capital. The people who use this service the most are young people, who are more active than older people, and they are very likely to stop by the supermarket to buy something for their trip.

In the traditional supermarket, the distribution shifts toward older ages, with a marked peak between 40 and 55 years old and a significant presence of customers over 60. This pattern indicates that this format attracts a more adult and established customer base, probably used to a stable shopping routine.

The convenience store has a more heterogeneous composition: on the one hand, there is a group of young customers, around 25–35 years old, who value proximity and speed of purchase; on the other, a group of adults aged 40–50 who also use this format as a complement to other channels.

Overall, considering all the supermarkets, the highest concentration of customers is between 25 and 60 years old, which shows that this age range is the core of consumption in the food retail sector.

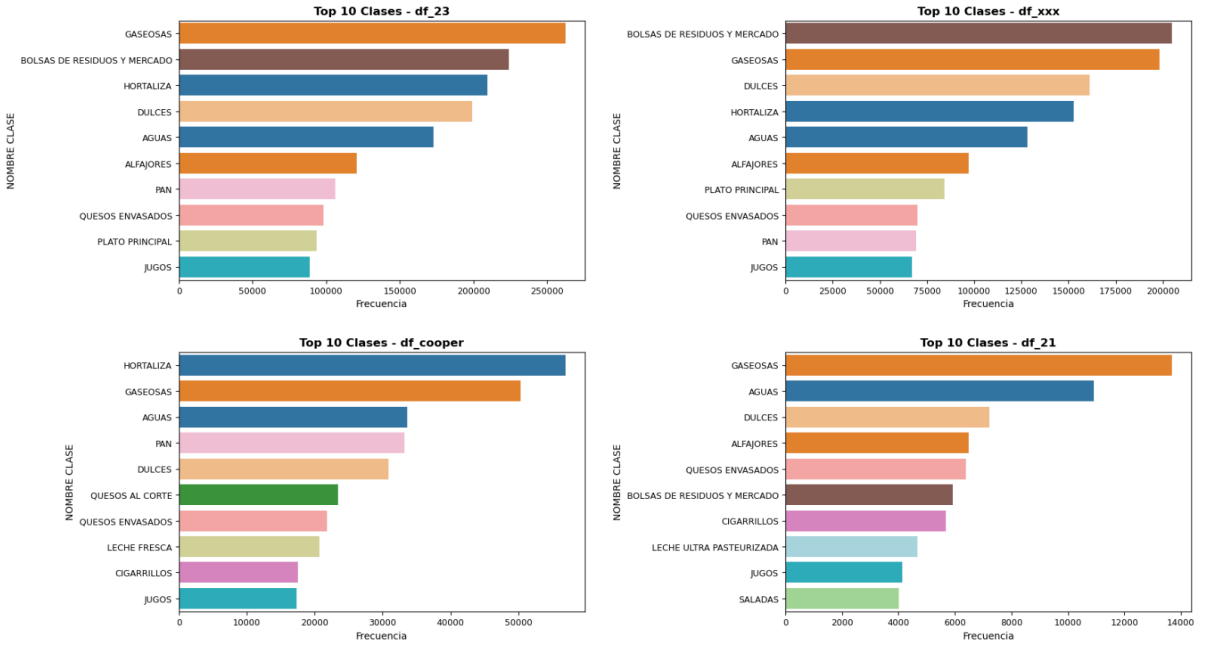


Figure 4.2.3.6 - Distribution of sales of the most frequent classes, for 2023, across the three stores studied and for each store individually

The analysis of the most purchased product classes shows clear differences between the different supermarket formats. In the aggregate total of all stores, categories such as soft drinks, garbage bags, vegetables, sweets and water stand out, reflecting a combination of products for immediate, everyday consumption together with basic, frequently replenished items.

In the case of the hypermarket, the behavior is similar to the overall pattern, although with a greater concentration in garbage bags and soft drinks, as well as a prominent presence of main dishes and packaged cheeses. This suggests that more planned, higher-volume purchases predominate in this format, typical of customers seeking to take advantage of assortment and bulk savings. The high share of soft drink and sweet sales is also associated with the bus terminal at Tres Cruces, since it is common to buy something to eat or drink during the bus trip.

The traditional supermarket, for its part, shows a greater weight of fresh products such as vegetables, bread and milk, along with categories such as deli cheeses and cigarettes. This pattern indicates that this format is more associated with replenishment and everyday purchases, where customers visit more frequently.

Finally, the convenience store has a different profile, with a clear predominance of soft drinks and water, followed by sweets, alfajores and cigarettes. In this case, consumption is characterized by impulse purchases and lower volume, focused on convenience and meeting immediate needs.

In conclusion, while hypermarkets concentrate more planned and larger purchases, traditional supermarkets and convenience stores are oriented toward replenishment and convenience purchases, respectively, showing how each format responds to different customer needs.

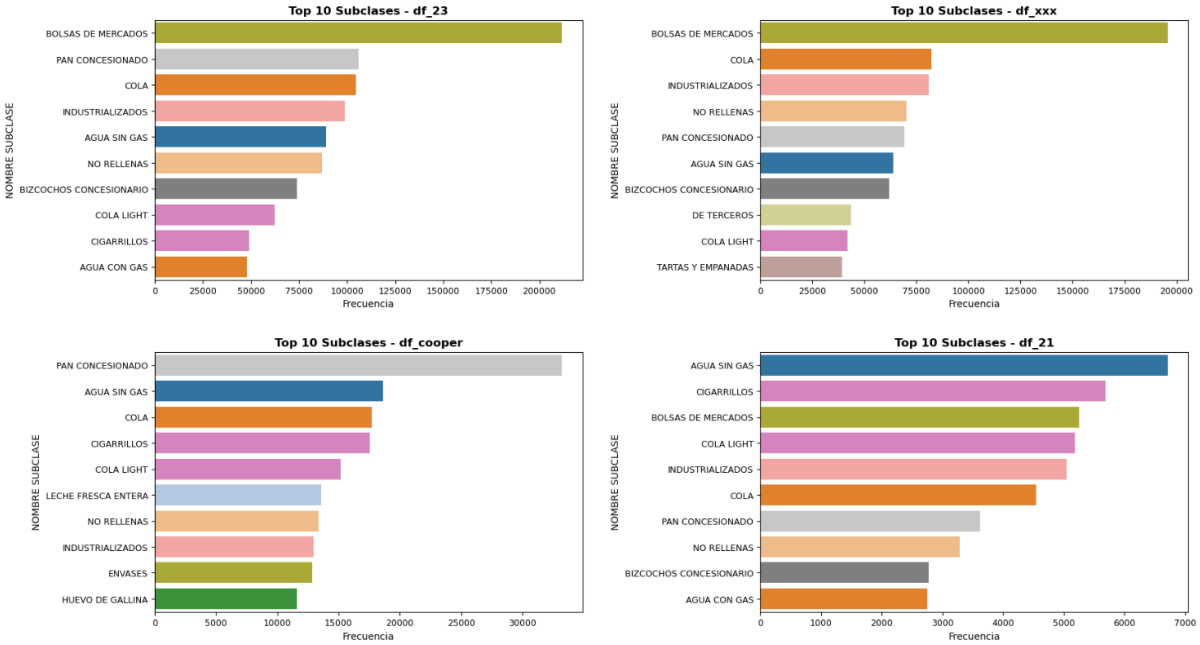


Figure 4.2.3.7 - Distribution of sales of the most frequent subclasses, for 2023, across the three stores studied and for each store individually

The analysis of product subclasses leads to a conclusion consistent with the patterns identified for the classes. The hypermarket mainly concentrates mass-supply items, such as bags, soft drinks and processed foods. The traditional supermarket is oriented toward fresh and everyday products, with bread, milk and eggs standing out. Finally, the convenience store shows a pattern associated with quick, immediate purchases, where beverages, cigarettes and pastries predominate.

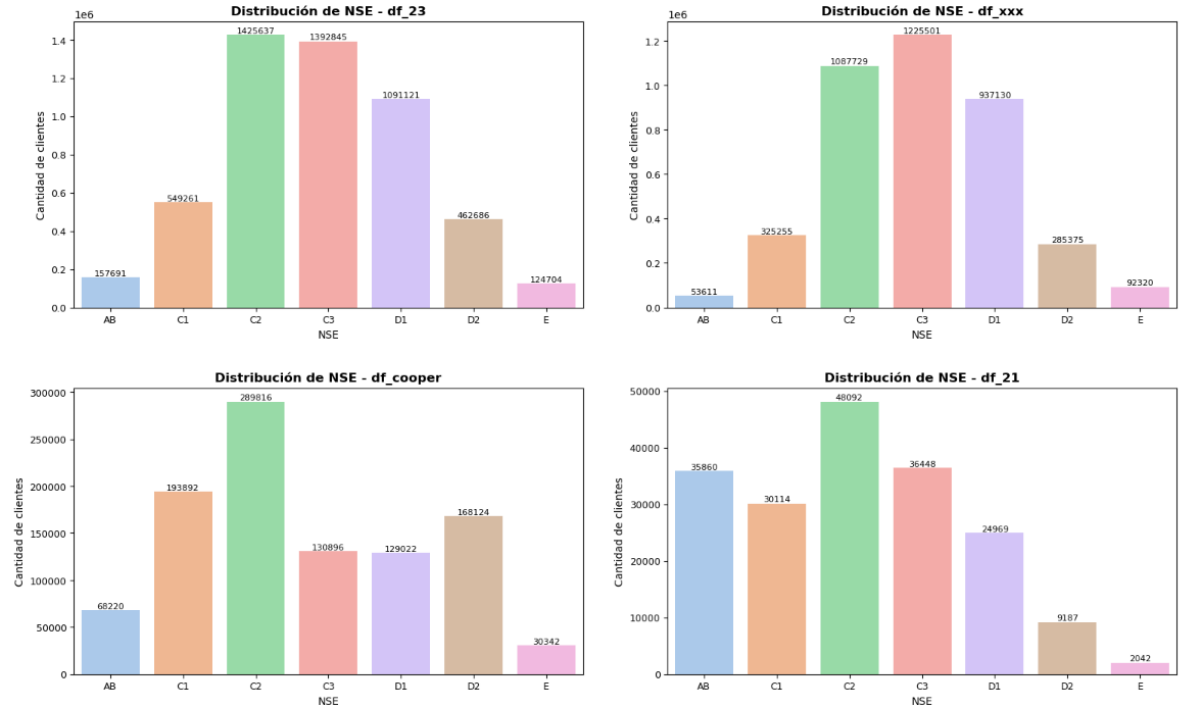


Figure 4.2.3.8 - Distribution of sales by socioeconomic level, for 2023, across the three stores studied and for each store individually

The analysis of the distribution by socioeconomic level shows that, across all supermarkets, the highest concentration of customers is in the middle segments, especially C2 and C3, followed by D1. In contrast, the extremes (the high (AB) and low (E) levels) represent a small proportion of the total. This confirms that the core of consumption in the sector is the middle and lower-middle class.

When looking at each format individually, the hypermarket shows a pattern very similar to the overall one, although with a relatively greater weight in the C3 and D1 segments. This behavior suggests that the hypermarket is more attractive to middle and lower-middle class sectors seeking to make large purchases and take advantage of more competitive prices. The distribution is consistent with its location. Being inside the country's main bus terminal, the store receives a very diverse daily flow of people traveling from different parts of Montevideo and the rest of the country. This means that individuals from all socioeconomic strata visit it, which is reflected in a broad and heterogeneous customer distribution. In this sense, it makes sense to consider that the composition of this hypermarket's customers is somewhat representative of the socioeconomic structure of the Uruguayan population in general, and approximates a normal distribution.

In the traditional supermarket, the predominant segments are still C2 and C3, but with a relatively greater presence of C1 and D2. This shows that this format serves a more heterogeneous public, ranging from the middle classes to lower-income sectors. The share of customers with a somewhat higher socioeconomic level can also be explained by the location of the store analyzed, in Carrasco Norte, a neighborhood with greater purchasing power than areas such as Tres Cruces, although not among the most exclusive in the city. In addition, the competition in this area includes supermarkets such as Tienda Inglesa and Devoto, which are more oriented toward a high socioeconomic audience, reinforcing the traditional supermarket's position as an intermediate alternative in the area.

Finally, the convenience store maintains the importance of the C2 and C3 segments, although the AB segment has a relatively greater weight compared to the other formats. This may be linked to the location of this type of store in diverse residential neighborhoods, where customers with greater purchasing power use them for convenience, even though their purchases are smaller.

In summary, all formats depend mainly on the middle class as the base of their clientele, although each shows nuances: the hypermarket is more oriented toward lower-middle sectors, the traditional supermarket concentrates a varied public, and the convenience store also manages to attract customers with greater purchasing power thanks to its proximity.

Regarding purchasing patterns, customer behavior was analyzed according to visit frequency. For this study, a frequent customer was defined as one who makes at least twelve purchases per year, equivalent to one visit per month. Based on this classification, only 8.5% of customers can be considered frequent, a considerably small percentage given the wide variety of products available in the chain. However, despite their low representation within the total customer base, this group accounts for 62.0% of the sales amounts of the stores analyzed and 62.9% of the number of invoices of the stores analyzed.

When comparing the data for 2023 and 2024, a remarkable stability in the general behavior patterns was observed. Both the most frequent classes and subclasses remained practically unchanged, as did the relative proportions between them. This consistency suggests that the structure of consumption and the customer types have stable dynamics over time, with no significant changes in the composition of demand.

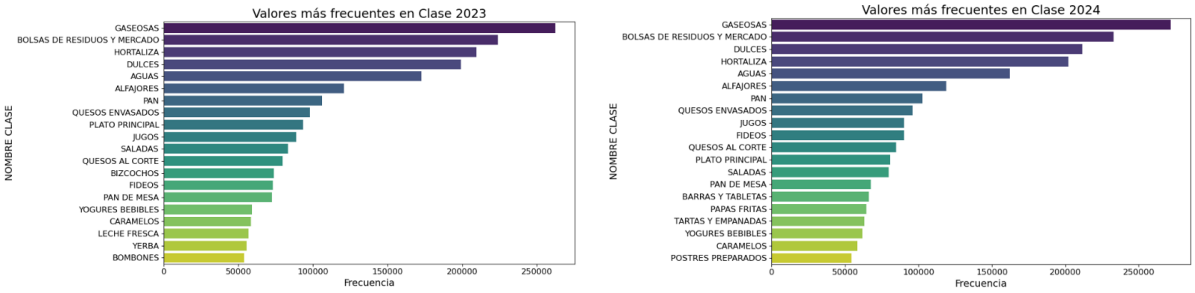


Figure 4.2.3.9 - Comparison of the most frequent classes between 2023 and 2024

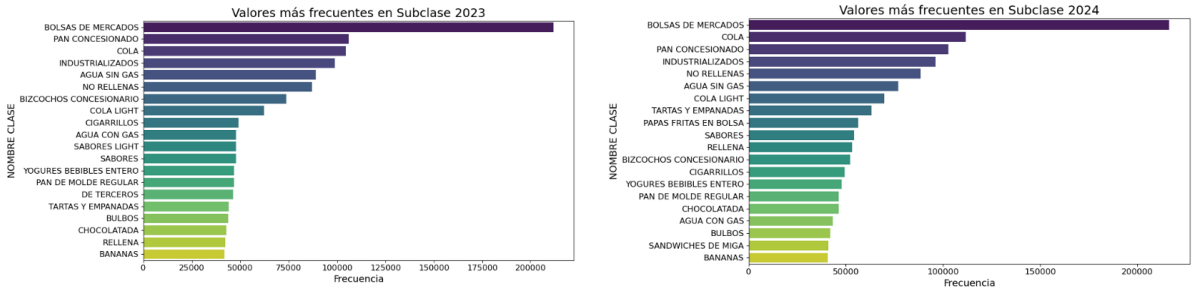


Figure 4.2.3.10 - Comparison of the most frequent subclasses between 2023 and 2024

Regarding the evolution of monthly sales, moderate variations were identified between the two years. The lower bound of sales increased from $48,000,000 to $52,000,000, while the upper bound remained stable at around $62,000,000. In both periods, the pattern of lower sales during the first two months of the year is repeated, followed by a marked increase in March. However, in 2024 the drop in April was more pronounced than in 2023, although an upward trend was later observed that brought monthly levels close to those of the previous year. In both years analyzed, September again stood out as a month of low commercial activity, but in 2024 there was a progressive rebound in October and November that had not been as evident in 2023. Finally, December was in both cases the month with the highest sales volume, showing an abrupt increase compared to November and the previous months.

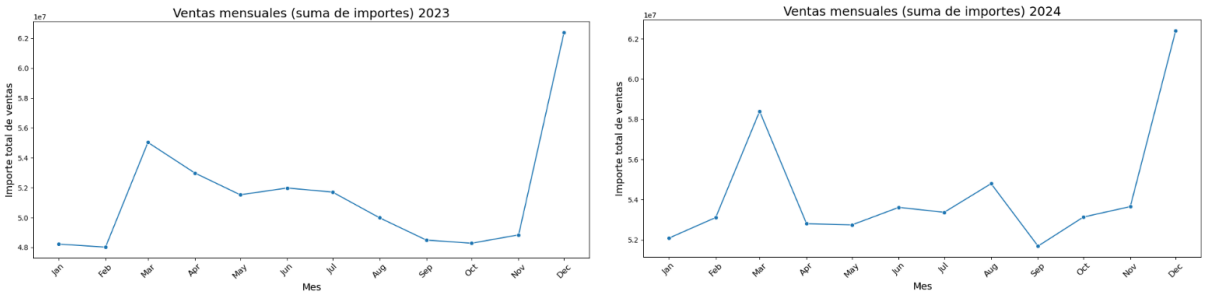


Figure 4.2.3.11 - Comparison of monthly sales amounts between 2023 and 2024

More marked variations were identified in the monthly average ticket. In 2023, values ranged between $110 and $135, while in 2024 this interval shifted upward, between $120 and $145. This difference reflects a general increase in the average purchase amount per customer, consistent with the upward trend observed in other indicators for the period.

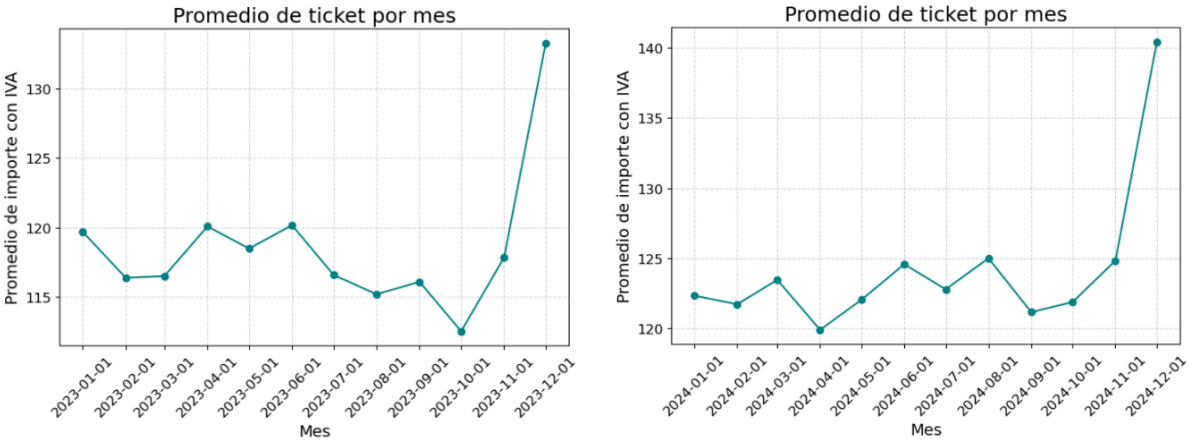


Figure 4.2.3.12 - Comparison of the average ticket by month between 2023 and 2024

4. 3 RESULTS AND MODELLING

Given the high level of noise in the transaction database and the size of the data volume analyzed, the dimensionality reduction and clustering methods applied did not produce consistent or statistically solid groupings. In this context, by noise we mean the combination of high distinctive variability in individual tickets (with many tickets containing very few items and showing rarely repeated product combinations), a weak signal of repeated patterns (that is, few consistent co-occurrences of product classes across tickets, which makes it difficult to identify stable groups) and the presence of atypical transactions, either because of very high or very low amounts or because of infrequent category combinations, which end up distorting the performance of the dimensionality reduction and clustering methods. In addition, the ticket–class matrix has a sparsity index of 99.72%, which indicates that almost all possible ticket and class combinations are empty cells, reinforcing the lack of repetitive structure from which the algorithms can learn. Given this methodological limitation, we chose to carry out a demonstrative clustering, aimed at illustrating the process and exploring the generation of soft consumer behavior profiles, focused more on showing the analytical potential of the technique than on obtaining definitive segments or robust conclusions.

The technique selected to generate the demonstrative profiles was based on a vector representation of the tickets using Word2Vec, complemented by a clustering process with K-Means. Word2Vec made it possible to transform the categorical product information into continuous numerical vectors, capturing co-occurrence relationships and association patterns between items that tend to be purchased together. To prevent the segmentation from being conditioned by the structure predefined by the company, the model was trained exclusively using the item identifier (ARTC_ARTC_ID), without the class and subclass variables.

A 128-dimensional vector space was defined, which determines how many numerical features are assigned to each product, seeking a balance between representational capacity and computational efficiency; a larger number would have allowed more complex relationships to be captured, but at the risk of increasing resource consumption and encouraging overfitting. The context window was set to 40 (Appendix 2), a value that indicates the number of nearby products within each ticket that the model considers when learning the relationships between products that appear together. This parameter was determined from the analysis of the distribution of the number of products per ticket, so that it coveredall or the vast majority of cases, preventing the model from being influenced by the order in

which products appear in the purchase record and ensuring that the captured relationships reflect real association patterns.

Regarding the results of the Word2Vec model, the learned vectors managed to capture intuitive relationships between products based on their co-occurrence in tickets. For example, in the case of tangerines, the cosine similarity analysis showed a value of 0.7349 with avocados, which is reasonable given that both belong to the same consumption season. Likewise, tangerines show values of 0.7455 with apples, 0.8447 with bananas and 0.8660 with oranges, all frequently purchased fruits associated with regular consumption patterns. These values show that the model correctly identifies natural groupings within the assortment, even without explicit information about categories or seasonality, relying only on the co-occurrence patterns observed in the transactional data.

The resulting vectors were used as input to apply K-Means in order to group tickets with similar purchasing patterns. The elbow method did not show a clear inflection point or make it possible to identify an optimal number of clusters. One possible hypothesis is that, due to the nature of the dataset, the inertia metric is notsuitable for the demonstrative techniques used at this stage. Consequently, the choice of six clusters was based on the previous analysis carried out with methodologies more appropriate for this type of data, which had shown consistent results before being discarded due to limitations related to computing capacity, the scale of the dataset and the level of noise observed. To improve model stability, n_init = 20 was set, since increasing it from 10 initializations did not produce substantial variations according to the Calinski–Harabasz metric, and it was adopted as a robust configuration. Finally, the max_iter parameter was set to 300, the default value used in scikit-learn, which is sufficient to ensure convergence without affecting the efficiency of the process.

The modeling, dimensionality reduction and clustering techniques evaluated and later discarded during the project were:

The possibility of applying an RFM (Recency, Frequency, Monetary Value) model was evaluated due to its wide presence in the literature and its frequent use in customer segmentation studies. However, the structure of the dataset prevented adequate results from being obtained. A considerable proportion of customers had made a single purchase, which generated a strongly skewed distribution and eliminated the model's ability to distinguish different behaviors. Under these conditions, the frequency variable no longer provides relevant information and, consequently, the recency metric loses its analytical meaning, since there is no repetition in purchasing behavior that would allow a recent customer to be interpreted as having greater value or a higher degree of engagement than another whose only purchase was made at the beginning of the period. In addition, the RFM approach requires analyzing the evolution of the metrics over time to reflect the recurrence and value of customer relationships. In this case, the size of the database, which covers several years of transactions, limited the possibility of carrying out this temporal tracking comprehensively, since processing multiple periods simultaneously exceeded the available computing capacity. For these reasons, we chose to explore alternative segmentation methods better suited to the type and volume of data, which would make it possible to capture complex relationships between products, tickets and customers without depending strictly on purchase frequency over time.

A combination of Word2Vec and DBSCAN was applied in order to generate a vector representation of the products in each ticket and then identify patterns through a density-based clustering method. This approach offered relevant theoretical advantages: DBSCAN automatically determines the number of clusters and explicitly identifies the points considered noise, which is useful in contexts characterized by high heterogeneity and the presence of outliers. However, the algorithm requires considerable

computing capacity, so it was necessary to apply undersampling techniques using scikit-learn's StratifiedShuffleSplit, reducing the sample to 25% of the dataset in order to run the model. Even so, the results were not suitable for the purposes of the project: after multiple parameter adjustments, the model classified approximately 88% of the records as noise and generated around 330 clusters from the remaining 12%. This type of output does not yield interpretable groupings or useful profiles for customer segmentation, so the technique had to be discarded.

Similarly, the implementation of an autoencoder was also considered as a tool to reduce dimensionality and obtain a more compact latent representation of the data. From a theoretical point of view, autoencoders make it possible to learn non-linear underlying structures within the data, preserving the most relevant relationships and removing noise. This would have made it possible to capture complex co-occurrence patterns between products, improving the subsequent clustering stage. However, applying the Autoencoder with DBSCAN proved computationally unfeasible due to the volume and dimensionality of the data. Both techniques have a considerably higher computational cost and require large amounts of memory and processing time, which exceeded the capacity available for the project.

A combination of undersampling, autoencoders and DBSCAN was also evaluated, with results very similar to the Word2Vec and DBSCAN approach: a high proportion of noise and an equally uninterpretable cluster structure.

In turn, a combination of undersampling, autoencoders and K-Means was evaluated. To apply K-Means, it was necessary to determine the optimal number of clusters, and k = 6 was identified using the elbow method. However, the resulting clustering had a silhouette score of 0.1, a value that shows little separation between groups and a poorly defined cluster structure. In the absence of a clear segmentation, we chose an approach that would cover the entire dataset and generate a representative clustering. Consequently, the Word2Vec and K-Means combination was finally implemented.

Initially, the plan was to evaluate cluster quality using the Silhouette index and to plot the results using the t-SNE (t-distributed Stochastic Neighbor Embedding) dimensionality reduction technique. However, due to the size and complexity of the dataset, these tools proved computationally unfeasible, as they required processing times and memory consumption well beyond the available resources. Given this limitation, we chose to use the Calinski–Harabasz index (CHI) and complement the analysis with other internal validation methods, such as the elbow method and exploratory tests with alternative algorithms, including DBSCAN. The results showed, on the one hand, that the CHI value increased as the number of clusters decreased (which indicates that the model fits better when the data is grouped into broad, less fragmented structures) and, on the other, that the values of the different validation criteria were low or barely differentiated across different K configurations. Likewise, DBSCAN failed to identify dense, well-defined groupings, and instead tended to classify large portions of the data as noise.

This behavior suggests that there are no clearly defined clusters within the dataset under the representations and methodologies used, and it is consistent with the presence of a high level of noise in the database. In this sense, when we state that the validation methods do not yield conclusive results, we mean that, from a strictly statistical perspective, no solid evidence was found to justify a robust and stable segmentation of tickets into a given number of clusters; that is, the clustering quality indicators do not support the existence of a strong segment structure or a "natural" partition of the data.

Nevertheless, and explicitly acknowledging this limitation, the approach was redefined toward a demonstrative or representative clustering. In particular, a value of K=6 was set, taken from the undersampling + autoencoder + K-Means attempt. This value of k is not the result of a statistical optimality criterion, but was chosen to illustrate the analytical potential of the technique and explore the generation of qualitative purchasing behavior profiles. Within this framework, the clusters obtained should not be interpreted as definitive segments or as a direct basis for decision-making, but as an exploratory approximation that makes it possible to identify soft patterns (for example, tickets more oriented toward replenishment, stock-up or mixed purchases).

This pattern can also be explained by the high heterogeneity of purchasing behavior, the noise inherent in transactional data and the absence of clear boundaries between customer profiles, factors that tend to favor models with fewer clusters and make it difficult to identify robust internal structures using algorithms such as K-Means.

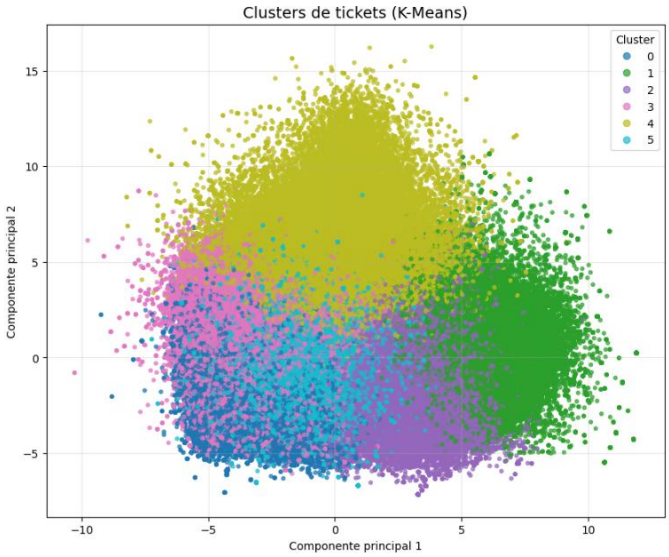


Figure 4.3.1 - Ticket clusters using K-Means on a 2D PCA representation.

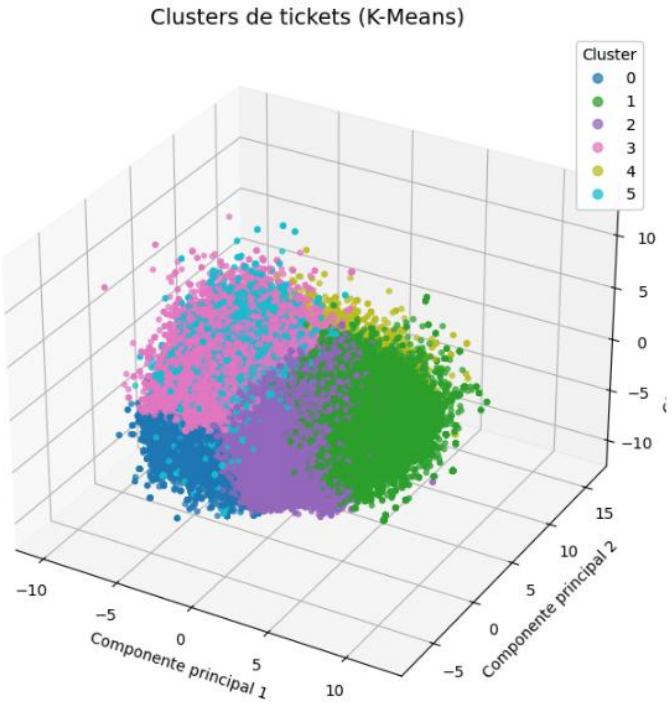


Figure 4.3.2 - Ticket clusters using K-Means on a 3D PCA representation.

CLUSTER ANALYSIS

Despite the limitations mentioned, especially the high level of noise and the fact that the statistical validation metrics do not solidly support the existence of well-defined clusters, the groups obtained through the demonstrative segmentation (K-means = 6) show patterns consistent with business logic and with common consumer behavior. This can be explained by the fact that, although the dataset does not show "strong" separations from a statistical point of view, it does contain soft signals linked to real purchasing habits which, although weak, emerge when a cluster structure is imposed. In this sense, although the exercise does not allow definitive conclusions to be drawn or operational segmentations to be built, it reveals an interesting tension: the results are statistically fragile but qualitatively revealing. Therefore, the analysis offers a valid first approximation of how groups of customers with differentiated behaviors could be organized.

When analyzing the proportion of customers who can be considered loyal within each cluster, the proportion of loyal customers is relatively homogeneous (Appendix 3). That is, even though some clusters are larger or smaller in absolute terms, the relative weight of loyal customers within each one remains stable. With the exception of clusters 0 and 3, the other groups show values between 6% and 9%.

Cluster 3 has the highest proportion of loyal customers (12.3%). This makes sense from a consumer behavior point of view: this cluster is mainly associated with fresh products (such as fruits, vegetables, meat, bakery products, etc.), categories where customer preference for a specific supermarket tends to be stronger, whether due to perceived quality, freshness, presentation or trust. In addition, for many consumers there is a real difference between buying fresh products at a supermarket and going to a street market, which further reinforces loyalty to the retailer when the experience is positive.

Cluster 0 also shows a relatively high proportion (11.8%). This proportion is consistent with the nature of this group. Since it is a cluster made up of everyday consumer products, customers tend to maintain

stable shopping routines, oriented toward weekly household replenishment. This type of behavior fosters loyalty, since the shopping experience in fresh and essential categories tends to create a preference for stores that guarantee quality, freshness, availability and predictable prices.

Charts used for the cluster analysis:

Main classes and subclasses of the clusters:

```
Cluster 0
CLASS_NAME
HORTALIZA                    89868
BOLSAS DE RESIDUOS Y MERCADO 60745
DULCES                       56823
FIDEOS                       49177
QUESOS AL CORTE              40510
LECHE FRESCA                 37929
GASEOSAS                     37356
PAN DE MESA                  33920
AGUAS                        32702
PAN                          30123
Name: count, dtype: int64

Cluster 1
CLASS_NAME
BOLSAS DE RESIDUOS Y MERCADO 17226
VESTIMENTA                   10198
PROTECCION FEMENINA           9043
CEPILLOS DENTALES             7065
DULCES                        6768
JABONES                       6538
GASEOSAS                      6248
MARCADORES                    6123
SHAMPOO                       5652
DESODORANTE MUJER             5636
Name: count, dtype: int64

Cluster 2
CLASS_NAME
GASEOSAS                     66282
AGUAS                        46385
PLATO PRINCIPAL              40810
TARTAS Y EMPANADAS           15245
SANDWICHERIA                 14107
ALFAJORES                    13980
BOLSAS DE RESIDUOS Y MERCADO  9463
JUGOS                         8698
BEBIDAS SABORIZADAS           7880
DULCES                        7281
Name: count, dtype: int64

Cluster 3
CLASS_NAME
HORTALIZA                    109408
BOLSAS DE RESIDUOS Y MERCADO  61180
QUESOS ENVASADOS              55650
AGUAS                         47987
DULCES                        39920
SALADAS                       39795
TROPICALES                    37296
GASEOSAS                      37056
PAN                           32462
CITRICOS                      32122
Name: count, dtype: int64

Cluster 4
CLASS_NAME
DULCES                       83279
ALFAJORES                    70843
GASEOSAS                     61372
BOLSAS DE RESIDUOS Y MERCADO 59014
BIZCOCHOS                    38752
CARAMELOS                    33615
BOMBONES                     28727
AGUAS                        25944
PLATO PRINCIPAL              25944
JUGOS                        24193
Name: count, dtype: int64

Cluster 5
CLASS_NAME
GASEOSAS                     54092
CERVEZAS                     28004
ENVASES                      18418
PAPAS FRITAS                 16904
BOLSAS DE RESIDUOS Y MERCADO 16402
PAN                          15710
AGUAS                        14863
CIGARRILLOS                  14833
SNACKS DE MAIZ               11069
OTROS SNACKS                  9987
Name: count, dtype: int64

Cluster 0
SUB_NAME
BOLSAS DE MERCADOS      57590
PAN CONCESIONADO        30123
LECHE FRESCA ENTERA     28339
PAN DE MOLDE REGULAR    26713
BULBOS                  24796
NO RELLENAS             23469
AZUCAR                  19055
JUGOS EN POLVO          18155
ARROZ BLANCO            18142
YERBA TRADICIONAL       18095
Name: count, dtype: int64

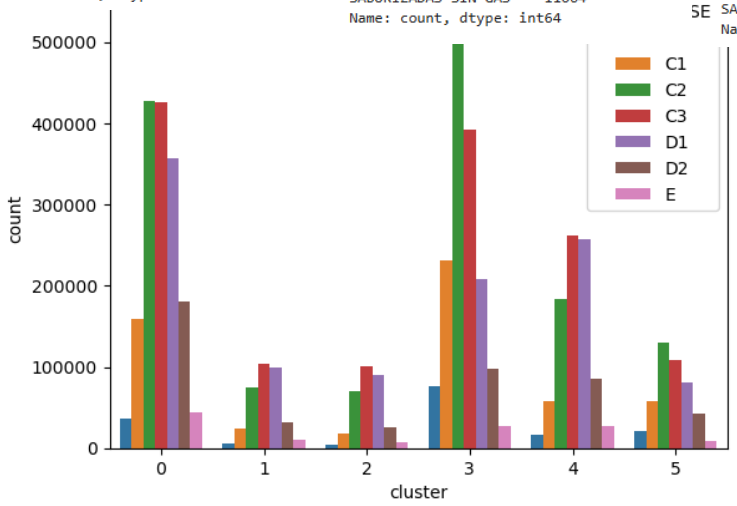
Cluster 1
SUB_NAME
BOLSAS DE MERCADOS              16232
CEPILLOS DENTALES ADULTOS        6206
TOALLAS HIGIENICAS               5483
JABONES COSMETICOS               5047
SHAMPOOS COSMETICO               3906
CREMAS DENTALES COMUNES          3775
DESODORANTES EN AEROSOL MUJER    3461
TOALLAS HUMEDAS COMUNES          3418
NO RELLENAS                      3317
DESODORANTES EN AEROSOL HOMBRE   3306
Name: count, dtype: int64

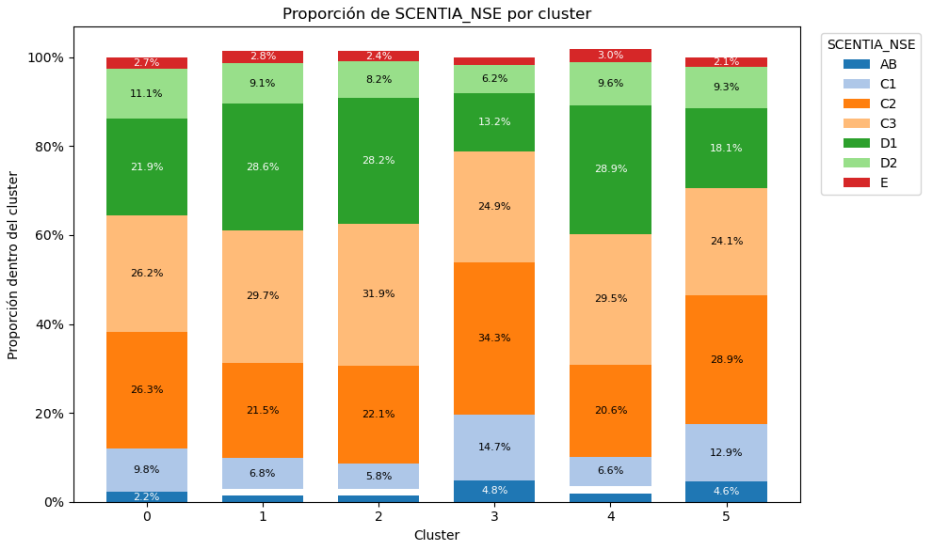
Cluster 2
SUB_NAME
COLA                    27783
AGUA SIN GAS            23146
DE TERCEROS             22228
TARTAS Y EMPANADAS      15245
COLA LIGHT              13815
SABORES                 12980
AGUA CON GAS            12131
SABORES LIGHT           11704
INDUSTRIALIZADOS        11392
SABORIZADAS SIN GAS     11064
Name: count, dtype: int64

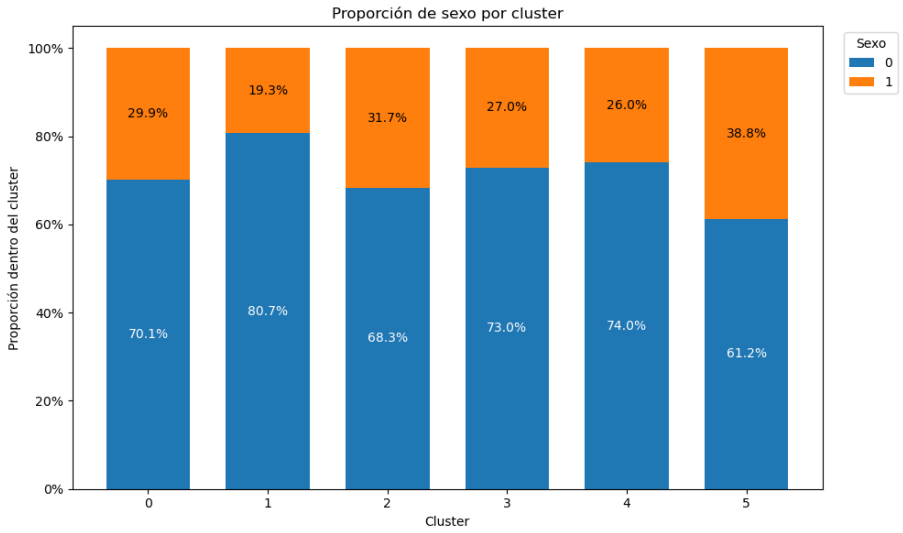
Cluster 3
SUB_NAME
BOLSAS DE MERCADOS      53886
PAN CONCESIONADO        32462
BANANAS                 29894
AGUA SIN GAS            29850
TOMATES                 23397
NO RELLENAS             20581
YOGURES BEBIBLES ENTERO 19418
FRUTOS                  18137
BULBOS                  17335
RAICES                  16898
Name: count, dtype: int64

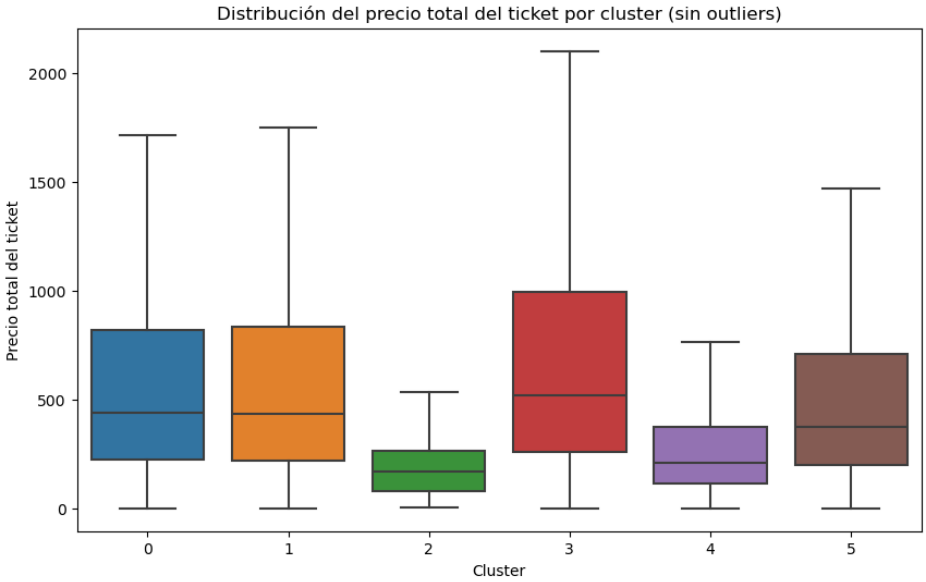
Cluster 4
SUB_NAME
INDUSTRIALIZADOS        61924
BOLSAS DE MERCADOS      58570
BIZCOCHOS CONCESIONARIO 38752
NO RELLENAS             34593
COLA                    29779
CHOCOLATADA             20995
PAN CONCESIONADO        20595
TARTAS Y EMPANADAS      20243
RELLENA                 20242
DE TERCEROS             15669
Name: count, dtype: int64

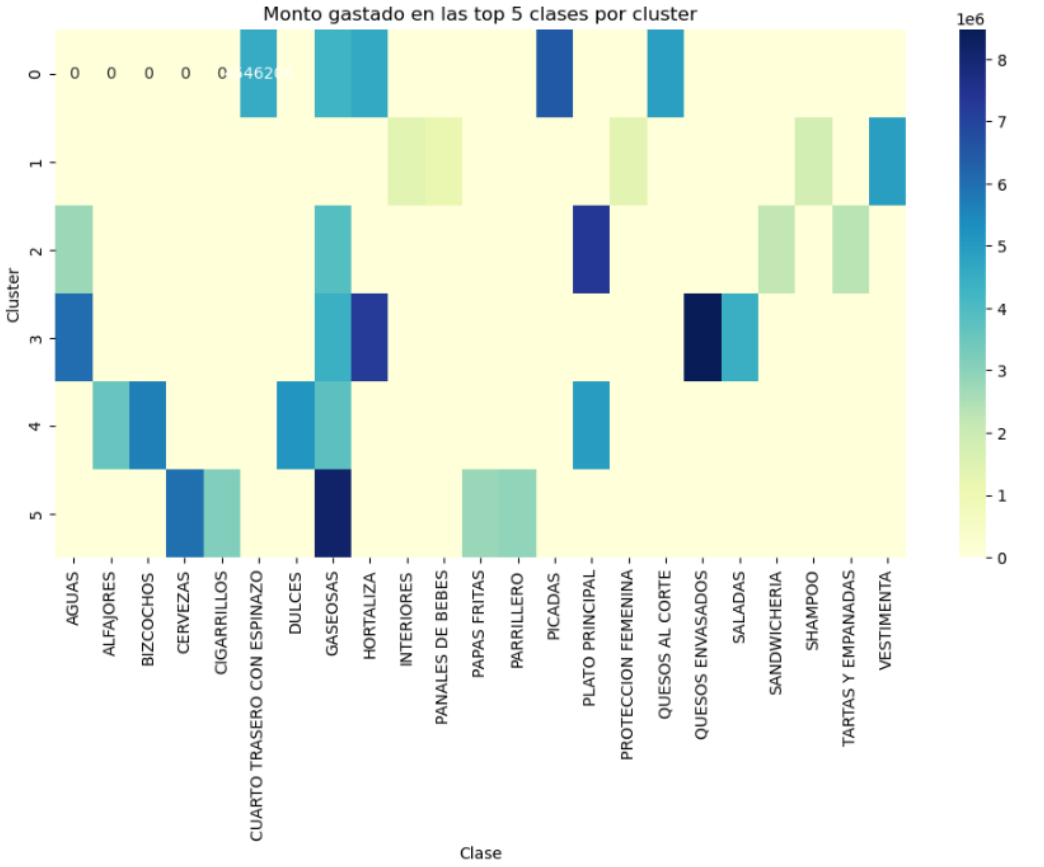
Cluster 5
SUB_NAME
COLA                    20768
ENVASES                 18418
CERVEZAS IMPORTADAS     16688
BOLSAS DE MERCADOS      16071
PAN CONCESIONADO        15710
CIGARRILLOS             14833
COLA LIGHT              13963
PAPAS FRITAS EN BOLSA   13160
CERVEZAS NACIONALES     10941
SABORES LIGHT           10411
Name: count, dtype: int64
```

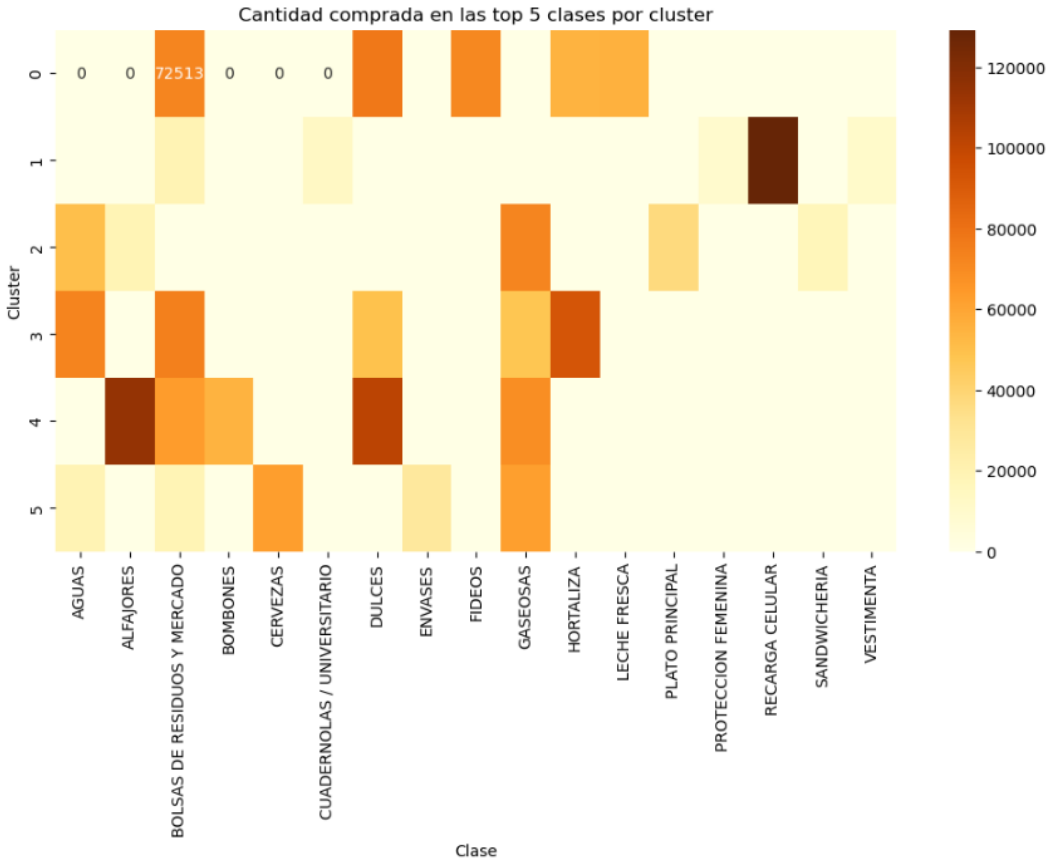


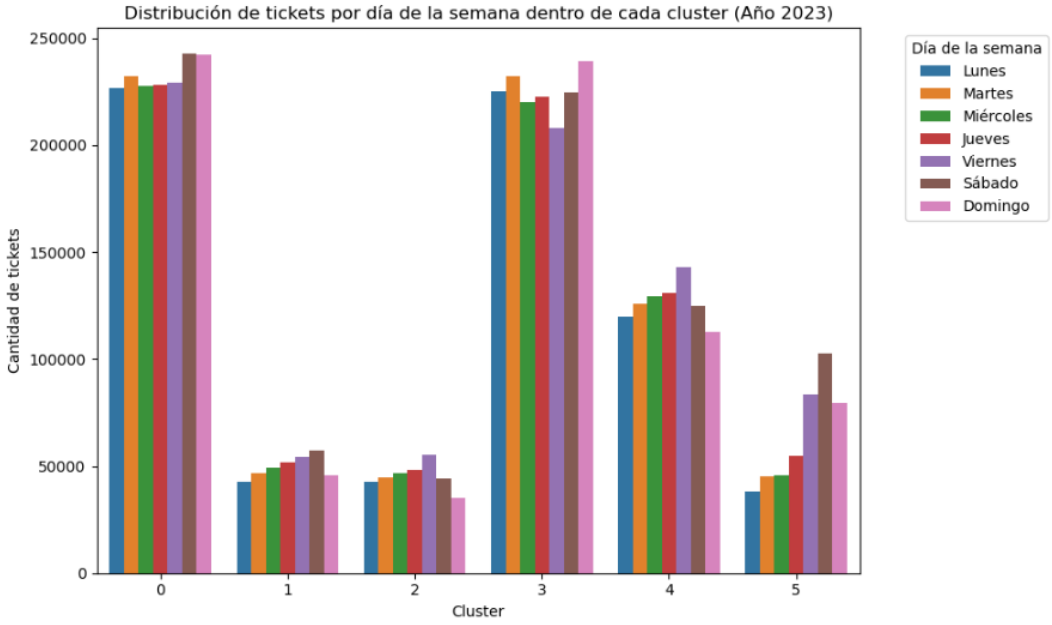

Proporción de SCENTIA_NSE por cluster
Proporción dentro del cluster
Cluster
SCENTIA_NSE
AB
C1
C2
C3
D1
D2
E
0%
20%
40%
60%
80%
100%
0
1
2
3
4
5
2.7%
11.1%
21.9%
26.2%
26.3%
9.8%
2.2%
2.8%
9.1%
28.6%
29.7%
21.5%
6.8%
2.4%
8.2%
28.2%
31.9%
22.1%
5.8%
6.2%
13.2%
24.9%
34.3%
14.7%
4.8%
3.0%
9.6%
28.9%
29.5%
20.6%
6.6%
2.1%
9.3%
18.1%
24.1%
28.9%
12.9%
4.6%


Proporción de sexo por cluster
Proporción dentro del cluster
Cluster
Sexo
0
1
0%
20%
40%
60%
80%
100%
0
1
2
3
4
5
29.9%
70.1%
19.3%
80.7%
31.7%
68.3%
27.0%
73.0%
26.0%
74.0%
38.8%
61.2%


Distribución del precio total del ticket por cluster (sin outliers)
Precio total del ticket
Cluster
0
500
1000
1500
2000
0
1
2
3
4
5


Monto gastado en las top 5 clases por cluster
Cluster
Clase
1e6
AGUAS
ALFAJORES
BIZCOCHOS
CERVEZAS
CIGARRILLOS
CUARTO TRASERO CON ESPINAZO
DULCES
GASEOSAS
HORTALIZA
INTERIORES
PANALES DE BEBES
PAPAS FRITAS
PARRILLERO
PICADAS
PLATO PRINCIPAL
PROTECCION FEMENINA
QUESOS AL CORTE
QUESOS ENVASADOS
SALADAS
SANDWICHERIA
SHAMPOO
TARTAS Y EMPANADAS
VESTIMENTA


Cantidad comprada en las top 5 clases por cluster
Cluster
Clase
72513
AGUAS
ALFAJORES
BOLSAS DE RESIDUOS Y MERCADO
BOMBONES
CERVEZAS
CUADERNOLAS / UNIVERSITARIO
DULCES
ENVASES
FIDEOS
GASEOSAS
HORTALIZA
LECHE FRESCA
PLATO PRINCIPAL
PROTECCION FEMENINA
RECARGA CELULAR
SANDWICHERIA
VESTIMENTA



Cluster 0 – Household and basic food purchases:

Cluster 0 is characterized by a concentration of everyday products, mainly vegetables, sweets, pasta, cheese, bread, fresh milk and soft drinks. This pattern suggests a type of purchase aimed at replenishing basic household food, probably made by consumers who plan their purchases. It is consistent that tickets from this cluster increase on Saturdays and Sundays, when it is common to do the shopping for the week.
The coexistence of fresh products (vegetables, bread, dairy) with processed foods (sweets, soft drinks, pasta) reflects comprehensive purchasing behavior, in which the customer seeks to stock up on a variety of essential products, combining staple foods with others for pleasure or occasional consumption.

In terms of needs, this cluster represents consumers who prioritize freshness, variety and affordable prices, making complete and planned purchases. It can be inferred that these are families or shoppers responsible for household supplies, given that the central 50% of the data is concentrated in individuals over 30 and under 60.

The distribution of socioeconomic level has an approximately normal shape, with a greater concentration in the C2 and C3 segments and a smaller presence at both the high (AB) and low (E) extremes. This structure is consistent with what would be expected given the overall composition of the supermarket's customers, where the middle groups usually represent the majority of regular shoppers. In this sense, the observed pattern suggests that Cluster 0 reflects a cross-sectional segment representative of the general public, with a balanced mix of consumers from different socioeconomic levels, but with a predominance of the middle sectors.

This behavior is consistent with the profile previously identified for this group, associated with household and basic food purchases, since these products are universal needs that attract consumers from different strata, maintaining a distribution proportional to the total customer base.

Cluster 1 – Quick consumption and convenience

Cluster 1 is dominated by products such as soft drinks, water, main dishes, sandwiches, pies and empanadas, alfajores and sweets, which suggests an orientation toward convenience or immediate-consumption purchases. The most frequent subclasses, such as main dishes, pies and empanadas, and quick meals, reinforce the idea of purchases associated with ready-to-eat meals, probably during working hours or while on the go.

In addition, as the week progresses from Monday to Saturday, the number of tickets belonging to this cluster increases steadily, which could be related to the growing fatigue accumulated over the working week. This pattern suggests that consumers tend to replace home cooking with ready-made or take-away options toward the end of the week.
This group represents a type of shopper who prioritizes speed, availability and practicality, possibly workers or students looking to sort out a meal or snack quickly. The predominance of non-alcoholic beverages and ready-to-eat food suggests purchases driven by the need of the moment rather than by planning..

In this sense, the cluster's main needs revolve around convenience, immediate access and variety in individual portions. It is the cluster with the fewest tickets.
From a sociodemographic point of view, this cluster stands out for having lower average ages than the other groups, concentrated between 25 and 40 years old, which may be attributed to the fact that young consumers tend to choose ready-made or prepared meals instead of cooking. This behavior pattern is aligned with more dynamic lifestyles, where speed and practicality prevail in purchasing decisions.

Regarding socioeconomic level, the distribution has an approximately normal shape, where the C2, C3 and D1 groups predominate, although with a slight skew toward the lower segments, with a greater proportion of consumers belonging to class D1. This shift suggests that the consumption of ready meals may also have an economic motivation, as it represents a low-cost, low-planning alternative to buying fresh ingredients or cooking at home.

Although women remain the majority within this group, there is a relatively higher proportion of men compared to most of the other clusters. This behavior could be linked to consumption patterns associated with work or study routines, where men tend to solve their daily meals with quick and accessible options.

Finally, the total price per ticket in this cluster is significantly lower than in the other groups. This indicates that purchases tend to be highly focused on specific products, such as ready meals, beverages or snacks, and do not include assortment or household replenishment items. Consequently, Cluster 1 reflects a pragmatic, young and convenience-oriented consumer profile, whose purchasing decisions are driven by speed, price, immediacy of consumption and "ready to eat" solutions rather than by planning or general stocking up.

Cluster 2 – Impulse consumption and indulgent products

Cluster 2 shows a high concentration of products such as sweets, soft drinks, alfajores, pastries, candy and chocolates, along with subclasses such as processed, filled and sweet snacks. This consumption pattern indicates impulsive or instant-gratification behavior associated with indulgent products.
The coexistence of sugary drinks and high-calorie foods reflects a consumer profile that seeks to satisfy cravings or small daily pleasures rather than make planned purchases. Likewise, the purchase frequency of this type of product tends to be high, although with low average tickets.

Compared to Cluster 1, this group shows very similar consumption behavior, although with a larger number of tickets and a slightly higher average amount. On the other hand, there is a slightly lower proportion of men than in Cluster 1.

In summary, Cluster 2 represents a need linked to pleasure, variety and impulse consumption. Consumers in this group prioritize immediate availability and promotions, valuing emotional gratification more than planning or unit price.

Cluster 3 – Fresh purchases and healthy eating

Cluster 3 mainly groups fresh and natural products, including vegetables, fruits (tropical and citrus), cheese, water, bread and yogurt. The most frequent subclasses, such as bananas, tomatoes, bulbs and roots, show a purchasing profile focused on perishable and healthy food. This behavior is associated with consumers who seek to maintain a balanced diet and who place special importance on product quality and freshness. The presence of drinkable yogurts and still water reinforces the idea of consumption oriented toward well-being, natural products and health.
In terms of socioeconomic level, this cluster has a greater concentration of customers from the highest strata (AB, C1 and C2) compared to the other groups. This pattern is consistent with empirical evidence indicating that consumers with greater purchasing power tend to prioritize fresh, natural and healthy products, associated with greater awareness of diet and nutritional quality. In this sense, healthy consumption can be interpreted as a reflection of the cultural and economic capital of the higher segments, who usually have both the knowledge and the financial means to access this type of product.
From an age perspective, Cluster 3 has the highest median age of all groups, with a significant concentration of consumers between 40 and 60 years old and a range extending toward older ages. This suggests the presence of an adult, mature audience with more stable, planned shopping habits oriented toward long-term well-being. This type of consumer tends to make more rational purchases, valuing quality over price, and maintaining a regular purchase frequency due to the perishable nature of the products purchased.
In line with the above, the analysis of total ticket price shows that this cluster has the highest tickets of the dataset, showing a higher average spend per purchase. This result is consistent with the profile of consumers from upper-middle and upper socioeconomic levels, whose purchasing behavior is characterized by lower price sensitivity and a greater willingness to pay for fresh, higher-quality products associated with a healthy lifestyle.
Regarding behavior throughout the week, purchases decrease toward Friday and increase markedly on Sundays, when the number of tickets peaks. This pattern suggests that consumers in this cluster tend to plan their fresh food purchases during the weekend, especially on Sundays, in preparation for the coming week. The Friday drop may be linked to the fact that many consumers postpone buying perishable products until the weekend, when they have more time to do more complete shopping and choose fresh products.
Overall, Cluster 3 represents a segment of mature consumers, from upper-middle and upper socioeconomic levels, who prioritize healthy eating, responsible consumption and freshness. It is a strategic group for designing premium offers, natural products and communication campaigns associated with well-being, where perceived value relies more on quality and experience than on price.

Cluster 4 – Hygiene, personal care and non-food items

Cluster 4 differs from the rest because its composition is centered on non-food products, mainly related to personal care and household hygiene. The most frequent classes include feminine care, toothbrushes, soaps, shampoo, deodorants and clothing, while the subclasses reinforce this trend with

items such as sanitary pads, toothpaste and aerosol deodorants. This pattern suggests a more planned, lower-frequency type of purchase, aimed at the periodic replenishment of personal or household items. In terms of volume, this group has one of the lowest numbers of tickets, together with Cluster 2, which matches the nature of the products it groups: replenishment and durable goods whose purchase does not require high recurrence.

From an age perspective, the age range of Cluster 4 is the most concentrated and narrow of all groups, with a median of around 40 years and relatively low dispersion. This suggests the presence of a young-adult and middle-aged audience, characterized by stable routines and predictable shopping habits. This age profile matches a stage of life in which consumption decisions tend to be more rational and function-oriented, reflecting conscious spending planning.

Regarding socioeconomic level, the distribution of this cluster resembles that observed in Clusters 0, 1 and 2, with a significant presence of levels C2, C3 and D1, which indicates a middle-income audience. This is consistent with the composition of the basket, which includes essential products purchased regularly by most households.

Regarding behavior by day of the week, purchases in this cluster increase progressively throughout the week, peaking on Fridays and then decreasing over the weekend. This trend suggests that purchases of this type of product are usually planned toward the end of the working week, when consumers set aside time to replenish household or personal items before the weekend. The distribution is similar to that observed in Clusters 2 and 3, reinforcing the idea of a structured and recurring purchasing pattern associated with planning rather than impulse.

In summary, Cluster 4 represents a segment of rational, planning-oriented consumers with middle purchasing power, focused on the functional and recurring purchase of hygiene and personal care items. It is a relevant group for brand loyalty strategies, repurchase programs and cross-promotions between complementary household categories.

Cluster 5 – Leisure, recreation and adult consumption

Cluster 5 is characterized by a prominent presence of beer, cigarettes, soft drinks, snacks (potato chips, corn snacks and others) and returnable containers. These categories reveal a consumption pattern oriented toward leisure and socializing, probably associated with recreational moments or weekends.

The combination of alcoholic beverages, cigarettes and side foods (snacks) suggests an adult consumer profile that makes purchases for moments of enjoyment or social gatherings. This cluster reflects less frequent purchasing behavior, but with a higher average ticket, linked to leisure situations.

Regarding socioeconomic level, this cluster has a distribution slightly skewed toward the higher levels, differing from most clusters and resembling Cluster 3, where consumers with greater purchasing power also predominate. This trend could be explained by the presence of products with higher unit prices or associated with premium brands, especially within the beverage and cigarette categories.

From a temporal point of view, weekly behavior shows an upward trend toward the weekend, peaking on Saturday, followed by a slight decline on Sunday. This pattern is consistent with the recreational nature of the predominant products in this cluster, since purchases tend to be concentrated on the days before rest or social activities. The lower activity during the first days of the week reinforces this interpretation, indicating that these are purchases planned for leisure rather than for everyday consumption.

The predominant needs in this group revolve around pleasure, convenience and the social experience of consumption, so commercial strategies based on bundled promotions or themed packs (for example, beer + snacks) could be particularly effective in attracting and retaining this segment.

5. POWER BI

DASHBOARD APPROACH

The dashboard was designed with a demand-oriented approach, prioritizing the creation of a practical and actionable tool that supports the chain's commercial and operational decision-making.

Unlike the analysis focused on customer behavior, the dashboard does not seek to study personal preferences or purchasing patterns at a micro level, but to understand the overall dynamics of sales across their different dimensions. The main objective was to provide an integrated and comparative view that makes it possible to analyze the evolution of demand by store, subclass, supplier, sales channel and store type, identifying structural differences between formats (such as hypermarket, traditional supermarket or convenience store) as well as relevant trends and seasonality.

In this way, the dashboard not only facilitates the detection of consistent patterns and the anticipation of changes in consumption, but also helps optimize supply and strengthen communication with suppliers. Together, these functions make it possible to achieve a deeper and more operational understanding of the chain's commercial performance.

DATA EXTRACTION AND MODELING

To build the dashboard, a direct connection was established between Power BI and the corporate Snowflake data warehouse, using the credentials provided by the company to each team member. The information was consumed from this environment because it is the official data layer prepared by the organization: there, the tables are integrated, curated and structured for analytical use, which makes it easier to incorporate them into Power BI and avoids working with raw data or data scattered across multiple sources. This integration made it possible to import the necessary tables securely and to work with real, up-to-date data, maintaining consistency with the organization's operational flows.

If the company decided to incorporate the dashboard into its production Power BI environment, it could schedule automatic refreshes using its own credentials, thus guaranteeing the periodic update of the indicators. The SQL queries used to generate the model are included in the appendices (see Appendix 4).

As mentioned previously, the chain's transactional volume made it impossible to work at a purchase-by-purchase level of detail without compromising performance. For this reason, we chose to build an aggregated transactional table, which made it possible to strike a balance between data representativeness, load speed and dashboard usability. This table summarizes the information by year and month (YEARMONTH format – YYYYMM), by store, by supplier and by product subclass. Each row reflects the monthly behavior of a specific combination of these dimensions, including both the number of units sold and the total amount sold in Uruguayan pesos. In this way, the two essential perspectives for demand analysis are preserved: physical volume and economic value.

During the construction of the tabular model, certain adjustments were also made following conversations with the company's team. Some stores and subclasses were deliberately excluded because they were considered to distort the reading of the data or to appear in the database for operational reasons that do not reflect real commercial activity. These decisions made it possible to clean up the model and ensure that the information presented in the dashboard more faithfully represented the day-to-day operation of the chain, preventing atypical elements from biasing the interpretation of the indicators.

To enrich the model and avoid duplicating information, each categorical dimension in the transactional table was linked to its respective lookup table through one-to-many (1:N) relationships. The lookup tables incorporated were the following:

- Suppliers, with their identifier, name and country of origin.
- Stores, including identifier, description, city, coordinates, geographic area (Montevideo or rest of the country), sales channel (digital or physical) and store type (hypermarket, supermarket or convenience store).
- Subclasses, with their code, name, the class they belong to and the type of sale (by unit or by weight).
- Classes, with their identifier and description.

In addition, the classes table is linked to the subclasses table through a 1:N relationship, reflecting the natural hierarchy of assortment categories: multiple subclasses belong to the same class, but each subclass reports to a single class.

Regarding time handling, the YEARMONTH column of the transactional table was linked to a monthly calendar table through a many-to-one (N:1) relationship. This calendar table defines, for each month, attributes such as the corresponding year, the quarter, the first day of the month and other variables relevant to the temporal segmentation of the analysis. In turn, this monthly calendar is linked to a daily calendar table through a 1:N relationship, preserving consistency between levels of temporal granularity and enabling future expansion of the model if the company decided to incorporate day-level analysis.

The end result is an optimized tabular model, capable of handling more than eight million records without compromising dashboard performance (a drastic reduction from the original 535 million), and structured so that demand can be analyzed flexibly by store, supplier, subclass, class, sales channel and time period. A screenshot of the data model is shown below.

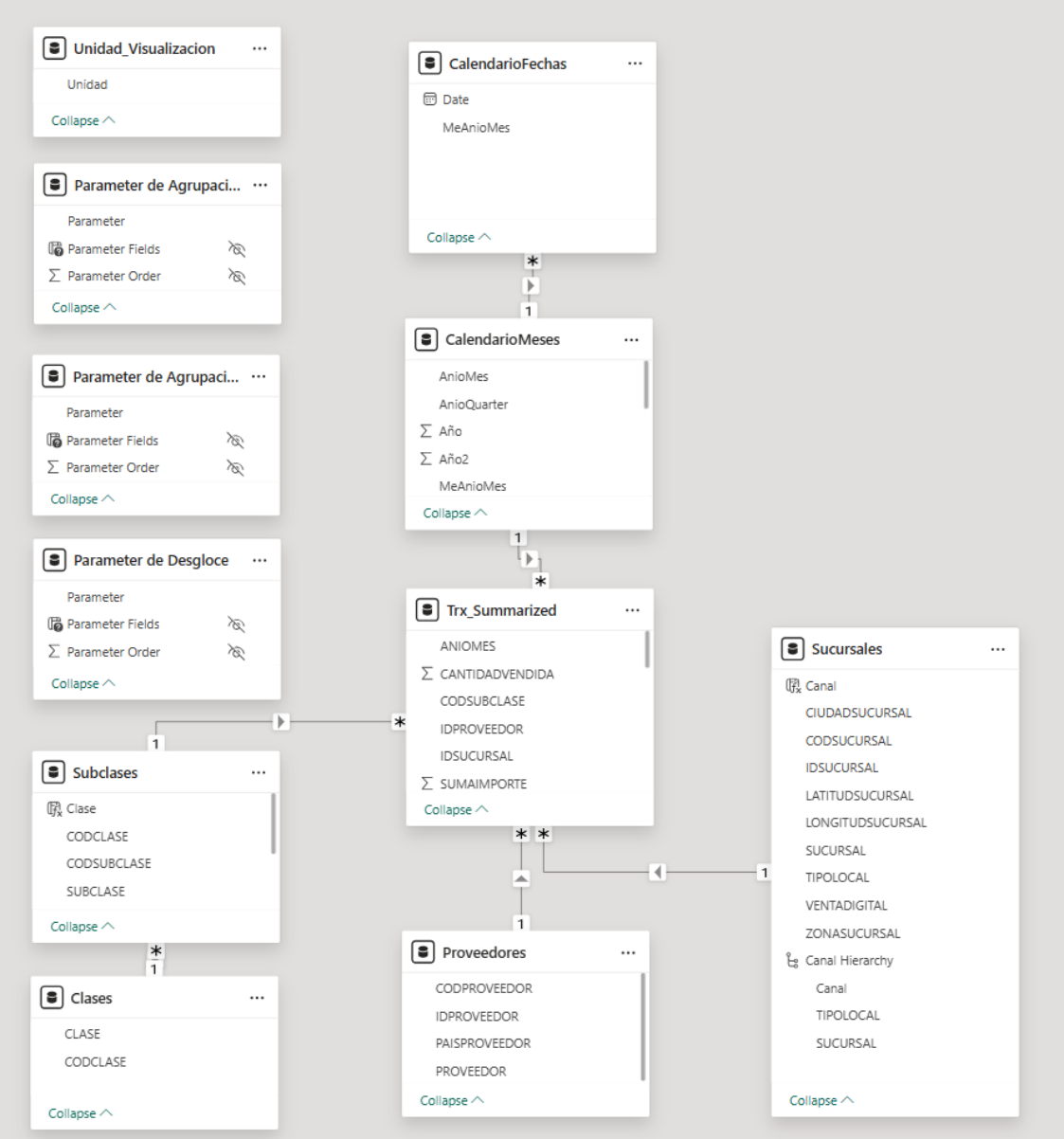


Figure 5.1 - Data modeling in Power BI.

The figure also shows certain tables that are not part of the main analytical model. These are auxiliary tables (such as parameters or structures used only as filters) whose purpose is to support navigation and the internal operation of the dashboard, but which do not contain transactional data or take part in the analytical joins of the tabular model.

PAGES

The dashboard was structured into four main pages, each designed to address a different perspective on demand behavior. This approach allows the user to explore the data progressively: from an overall view of the business to detailed analyses by specific category. All pages share a consistent set of filters that enable consistent comparisons across time units, channels, types of sale and store types, maintaining the dashboard's analytical logic.

The organization of the pages responds to two central objectives:

1. to offer an integrated reading of the evolution of the chain's sales
2. to make it possible to drill down into particular categories (subclasses, stores and suppliers) to identify relevant differences in demand behavior depending on the selected angle of analysis.

To ensure consistent interpretation across all pages, the dashboard incorporates two mandatory filters that structure the reading of the analysis. The first allows the user to choose whether the visualizations should be shown in physical units or in amount, which makes it easier to examine demand both from the perspective of volume sold and of its economic value. The second filter requires choosing the time unit used to represent the evolution series (month, quarter or year, among other options), allowing the temporal granularity to be adapted to the needs of the analysis. Combining both filters gives the user the ability to interpret the data from different angles and ensures that all visualizations remain consistent with each other.

Beyond the visualizations shown in this document, it is important to note that the value of the dashboard does not lie only in its final appearance, but in the technical planning behind it. The model structure, the efficiency of the relationships and the definition of the measures were intentionally designed to balance performance, clarity and business usefulness. As a result, a significant volume of information (three years of sales across more than one hundred and twenty stores) can be analyzed smoothly and without delays. Moreover, despite working with already aggregated data, without the level of detail of the individual purchase, the specific product or the end customer, it was possible to build a tool capable of extracting valuable and, above all, actionable information. The decisions made around the mandatory filters, the dynamic measures and the organization of each page aimed to offer a clear and consistent experience that makes the most of the available information without adding unnecessary complexity.

Overview page: Sales evolution

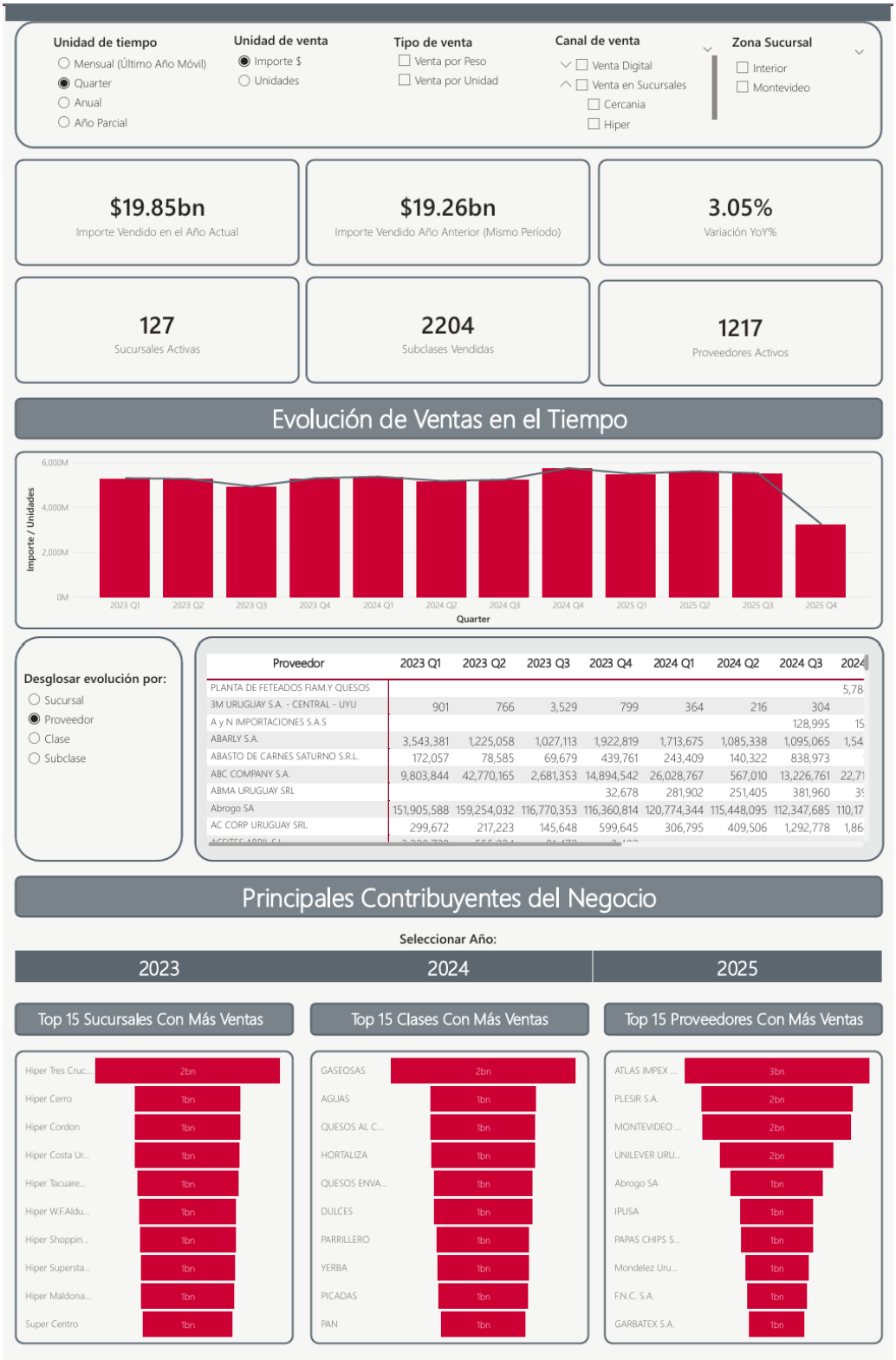


Figure 5.2 - Dashboard page 1.

The first page of the dashboard offers an overview of the chain's commercial performance, clearly showing how sales evolve over different periods. Its purpose is to provide a starting point for the analysis, making it possible to quickly see how demand behaves at the aggregate level and how it varies depending on the selected time horizon (month, quarter or year).

At the top, there are several cards with simple indicators summarizing the main values for the period: current-year sales, sales for the same point in the previous year, year-over-year change, and the number of active stores, subclasses and suppliers. Although the dashboard allows detailed and disaggregated analyses, it is essential that it also offers a clear and accessible initial reading. These cards serve that purpose by providing an overview of commercial performance, allowing the user to quickly get an idea of the state of the business before diving into more complex visualizations.

Next, there is a bar chart showing the evolution of sales over time and, next to it, a matrix whose level of detail can be selected from a dedicated filter. This makes it possible to break down the evolution by store, supplier, class or subclass, combining the overall reading with a more specific analysis of the segments that explain the variations in demand.

The page is complemented by three visualizations showing the top 15 stores, subclasses and suppliers by sales volume. These rankings help gauge the relative weight of each category within the total and make it possible to quickly spot the most relevant players. In addition, this section includes a dedicated filter to select the year for which the rankings are displayed. In this way, the user can see how relative positions change from one year to another, identify categories that gain or lose prominence and analyze the evolution of the main contributors over time.

All these visualizations respond fully to the filters available on the page, such as type of sale (by unit or by weight), channel (digital or stores) or geographic area.

Second page: Analysis by channel and store

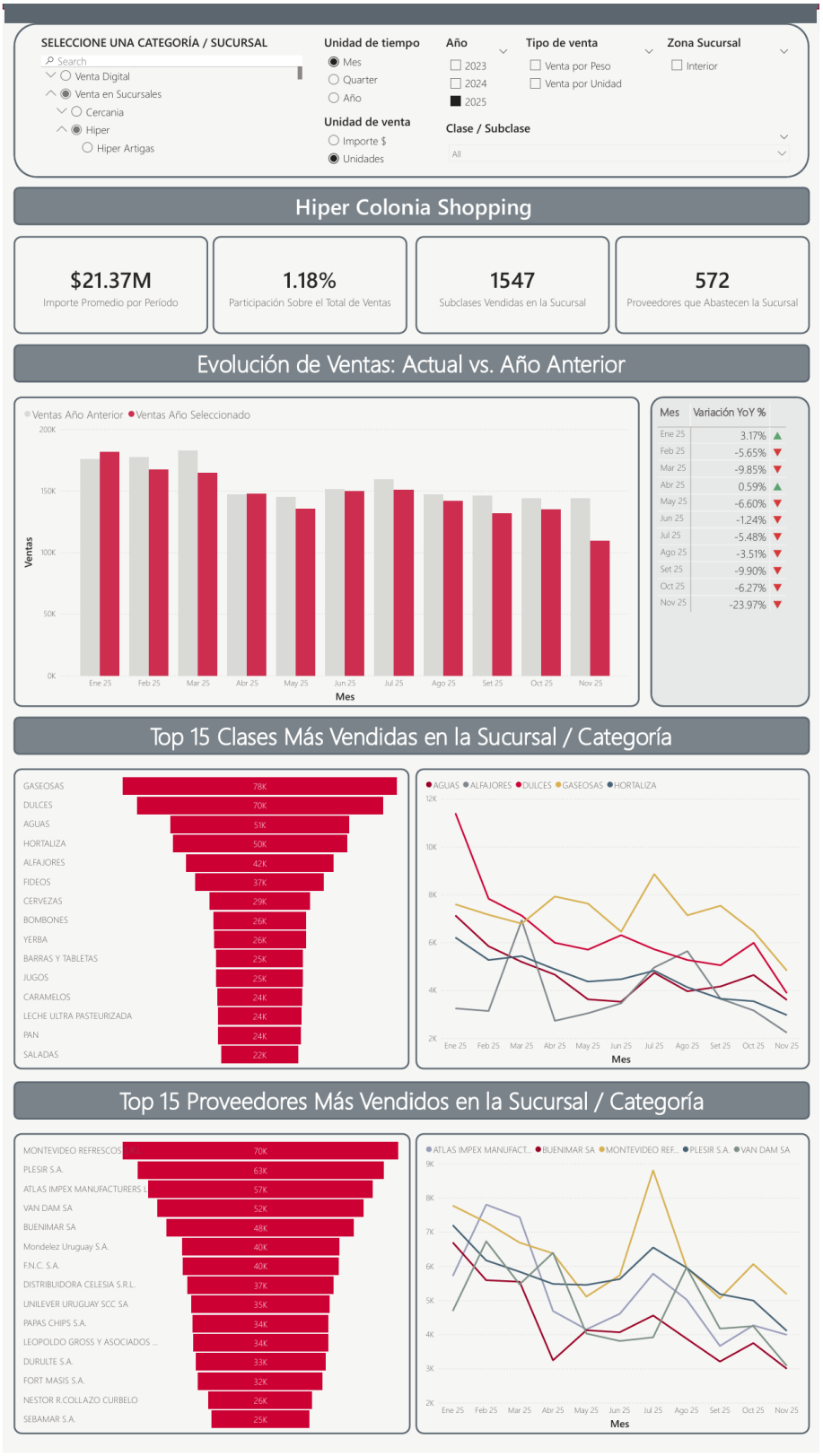


Figure 5.3 - Dashboard page 2.

The second page makes it possible to analyze the behavior of a specific store (or of a group of stores, when the user selects store types or broader categories) from a local perspective. This page aims to

identify which products drive the store's sales, how its sales evolve and what role the store plays within the chain's total.

After selecting the store (or category), indicators summarizing its recent performance are shown. These include the amount and quantity sold, averaged according to the selected period (monthly, quarterly or yearly), its share of total sales, the number of subclasses sold and the number of suppliers that supply the store. These indicators make it possible to quickly gauge the size, diversity and relative relevance of the store within the chain.

The main visualization on this page compares the evolution of sales over time with the same period of the previous year. This shows changes in the store's behavior, making it possible to distinguish between one-off episodes and more persistent trends.

The page also includes visualizations that combine a ranking with a temporal perspective. On the one hand, it shows the subclasses with the highest sales volume in the store, and within that set it highlights the evolution over time of the top five, which makes it possible to identify which categories explain most of the store's performance and how they vary over time. Similarly, it shows a ranking of the most relevant suppliers for the store, complemented by a chart showing the evolution of those with the greatest relative weight. This combined approach makes it easier to understand not only who the most important players are, but also how their sales behave over time.

Third page: Analysis by supplier

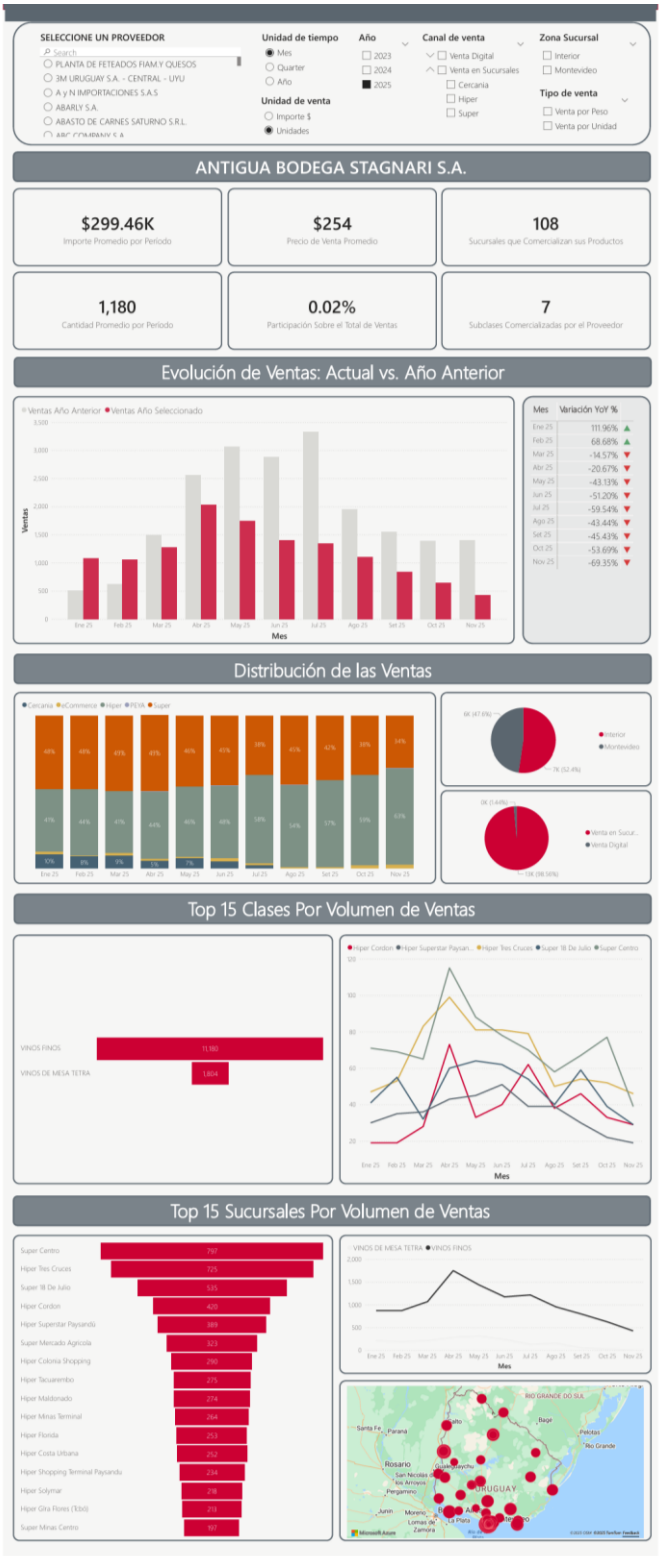


Figure 5.4 - Dashboard page 3.

The third page analyzes the performance of a specific supplier within the chain, making it possible to assess its contribution to the business, how its products are distributed across the different stores and which categories are most relevant in its portfolio.

The indicators at the top summarize the supplier's overall performance: amount and quantity sold, averaged according to the selected period (monthly, quarterly or yearly), the average selling price, its share of total sales, the number of stores where it has activity and the number of subclasses it sells. This data makes it possible to quickly gauge the supplier's importance within the assortment.

As on the previous pages, there is a temporal comparison between the current year and the previous year, making it easier to detect fluctuations in demand. There is also a visualization of the distribution by channel and store type, which makes it possible to identify whether the supplier has a greater presence in physical stores, in the digital channel or in certain store formats.

There are also visualizations that make it possible to analyze both the subclasses and the stores most relevant to the supplier. In both cases, there is a ranking of the categories with the highest sales volume together with a chart showing the evolution over time of the most prominent ones. Finally, an interactive map complements this analysis by showing in which stores its products have the greatest presence, represented by circles proportional to the level of sales.

Fourth page: Analysis by Class and Subclass

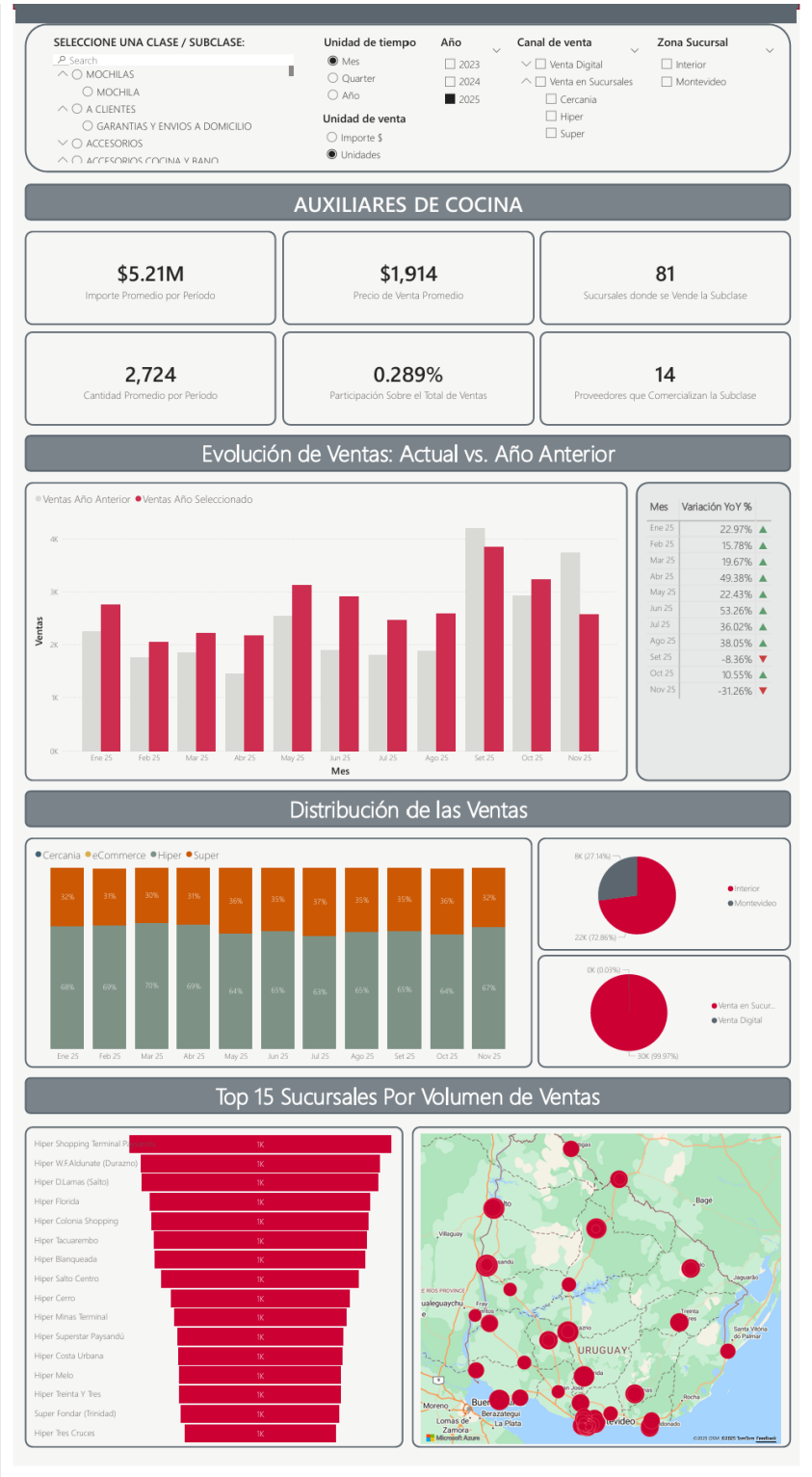


Figure 5.5 - Dashboard page 4.

The fourth page is aimed at the detailed analysis of a specific product category. Here, the user must select a subclass or class to access a focused reading of its behavior in the chain. The objective is to understand how demand for that particular category evolves, what its weight is within total sales and in which contexts (stores, channels or formats) it becomes most relevant.

The page begins with a set of indicators summarizing the performance of the selected category. These include the amount and quantity sold, averaged according to the chosen period (monthly, quarterly or yearly), the average selling price, the number of stores where the category records sales, its share of the total and the number of suppliers that sell it. These indicators facilitate a quick reading and make it possible to gauge the relative importance of the category within the chain.

Next, there is a comparative chart between current sales and those of the same period of the previous year, which makes it possible to assess recent variations and detect signs of growth or decline. This component is especially useful for categories sensitive to seasonality.

Complementing this analysis, the page includes visualizations showing the distribution of demand by sales channel, store type and type of sale, which makes it possible to detect structural differences in the category's behavior. It also includes an interactive map that represents each store in the country with a point proportional to the level of sales, making it easier to identify geographic patterns and regional differences.

Finally, the page shows the Top 15 stores where the selected subclass sells the most, accompanied by a line chart showing the evolution over time of the Top 5. In this way, the page makes it possible to analyze in detail where demand is concentrated and how it varies over time across the main points of sale.

MEASURES AND METRICS USED

The measures developed in Power BI were designed so that the dashboard could adapt to different ways of analyzing demand, without duplicating visualizations or creating parallel versions of each page. To do this, a limited set of measures was defined that make it possible to switch between units of analysis (amount or units) and between different time scales, always maintaining consistency between visualizations.

The core measure of the dashboard is _Ventas_Dinámico, which serves as the basis for most calculations. This measure responds directly to the Unit of sale filter: when the user selects "Units", it sums the quantity sold; when they select "Amount $", it sums the amount sold. In this way, all the pages of the dashboard can be analyzed from both the physical volume and the economic perspective. This duality is essential, since demand behavior does not always coincide with economic contribution: what sells in large quantities is not necessarily what generates the most revenue, whether at the level of categories, stores or suppliers.

In addition, a field parameter was incorporated to manage the selection of the time level at which the evolution of sales is analyzed. This makes it possible to view trends by month, quarter, full year or partial year, the latter option being available only on the overview page of the dashboard. This view is especially useful because it avoids distorted comparisons when the current year has not yet ended: each historical year is cut off at the same point in time as the current year, which makes it possible to assess how demand evolved in previous years at the same point in the calendar. To implement this behavior, the _Ventas_Mix measure was created, which combines the logic of _Ventas_Dinámico with the time parameter, integrating both dimensions into a single, consistent calculation.

On this first page, the user can switch between the yearly evolution of the last two years, the quarterly evolution for the same period, the monthly evolution of the current year or the partial-year view. This

flexibility makes it possible to adapt the reading of overall behavior to the type of analysis to be carried out.

On the other pages of the dashboard, the time selection is limited to monthly, quarterly or yearly scales. The user can also narrow the horizon by selecting a specific year, which helps keep the visualizations clear when the full period covers too long a span.

To enable comparative analyses, additional measures were included to contrast current performance with that of the previous year. These include _Ventas_Dinámico_PrevYear, which captures sales for the same period of the previous year, and _Ventas_Dinámico_YoY%, which calculates the year-over-year percentage change. These measures make it possible to assess whether the company (or the category analyzed) is doing better or worse than at the same point in the previous year, identifying relevant changes in demand behavior.

The dashboard also incorporates a set of measures designed to feed the cards used as KPIs on all pages. These include:

- the number of active stores or, depending on the case, the number of stores in which the selected subclass or supplier's products are sold;
- the number of subclasses sold in the store, or the number of subclasses sold by a supplier;
- the number of suppliers that supply the store or subclass analyzed;
- the share of the selected category in total sales;
- specific measures for subclasses and suppliers, such as total units sold, total amount sold and the resulting average price.

Finally, the dashboard model incorporated the use of hierarchies, a feature that makes it possible to organize categories into different levels within the same field and facilitates navigation between scales of analysis without having to change filters. For products, a hierarchy was built that makes it possible to switch between classes and subclasses within the same selector, which simplifies exploration of the assortment and allows the level of detail to be adapted to the needs of the analysis. Similarly, a hierarchy was defined for stores that organizes the different formats and sales channels, making it possible to analyze demand from a more general or more specific perspective. This approach reduces the number of filters needed and contributes to a more intuitive navigation of the dashboard.

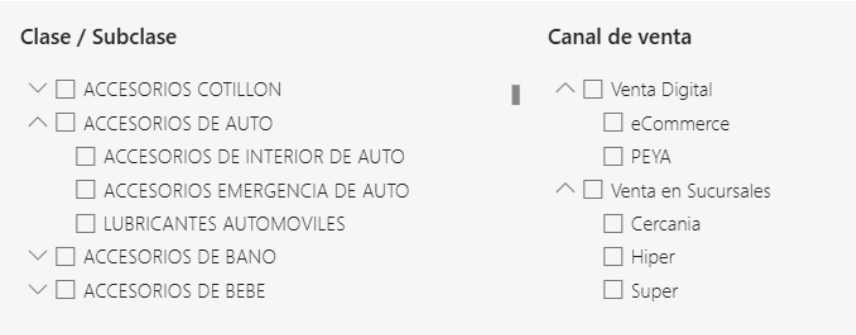


Figure 5.6 - Hierarchies.

The measures and parameters mentioned are presented in detail in Appendix 5.

6. GENERAL CONCLUSIONS

The project made it possible to transform a massive and complex dataset into structured and analytically useful information, providing a deeper understanding of the general consumption patterns in the chain. Through the combined use of vector representation techniques (Word2Vec) and clustering algorithms (K-Means), a soft clustering was generated, aimed at exploring possible ways of structuring customer profiles based on their purchasing behavior. Although a definitive or operational segmentation was not obtained, the results reflect subtle differences that are consistent with business logic and human behavior, offering a valid first approximation of how groups of customers with different consumption habits could be organized.

The experience showed the importance of correctly defining objectives and scope when working with large-scale databases. Computational limitations, the heterogeneity of consumer behavior and the broad definition of the concept of "loyalty customer" highlighted the need to use analytical approaches that are scalable, adaptive and suited to the volume and quality of the available data. The analysis also showed that the current loyalty program does not necessarily distinguish customers who are truly attached to the brand from those who use the card only occasionally. This opens the door to reviewing both the definition of loyalty and the associated incentives, in order to improve their effectiveness and alignment with actual consumption patterns.

From a methodological perspective, the project showed that hybrid approaches, such as combining embeddings and clustering techniques, are especially promising in retail environments with high dimensionality and unstructured data. It also highlighted the importance of working with cleaner, more consistent and more complete data, especially regarding the loyalty program, to raise the interpretive quality of future models.

The results illustrate the analytical potential of these techniques, but also reveal the limitations imposed by the available infrastructure and data quality. With a more solid base, it would have been possible to evaluate the evolution of the behavior of thousands of customers over time, detect transitions between purchasing patterns and build a more granular dynamic segmentation, capable of anticipating churn risk and informing commercial decisions more proactively.

The development of the Power BI dashboard complemented the previous analyses by offering a structured visualization of demand, making it possible to observe its evolution across different time scales and compare performance across stores, subclasses and suppliers. Its design, based on aggregated data and a flexible filter structure, makes it easier to identify trends and seasonality, recognize differences between store formats and sales channels, and detect relevant variations that may influence commercial planning. Although the dashboard does not seek to analyze individual customer behavior (as the clustering does), it provides a valuable operational perspective on the overall dynamics of the business, enabling continuous sales monitoring and supporting decisions related to assortment, supplier management and anticipating changes in demand.

Future recommendations:

- Based on these observations, we suggest using these demonstrative clusters as a starting point to redesign the loyalty program, directing incentives toward behaviors that reflect true loyalty and not just registration in the system or sporadic use of the card. At a strategic level, one concrete line of action is to design a differentiated benefits scheme according to the identified profile, where customers with the highest potential value receive incentives aligned with their behavior (specific discounts, personalized benefits, relevant content), while lower-engagement segments are encouraged through actions aimed at increasing their frequency and average

ticket. In this way, the loyalty program could move from a mainly transactional model to a strategic tool for managing customer relationships, reducing the proportion of inactive users and strengthening recurrence. It should be noted that, during the project, the company's team stated that it is already working on distinguishing between recurring and loyalty customers, so these results should be understood as a complementary input to the initiatives currently under development.

- In addition, the need to improve the quality and internal structure of the data was identified, particularly regarding the definition and organization of the assortment. We recommend reviewing and cleaning up the classes and subclasses, since the current structure has heterogeneous and, in some cases, redundant levels of granularity. A more coherent reorganization would allow classes to represent broad and conceptually clear categories, while subclasses, although specific, could be optimized to avoid unnecessary dispersion. A tidier structure would facilitate interpretation, make it possible to identify trends more precisely and strengthen the analyses carried out in both modeling processes and the Power BI dashboard.
- Along the same lines, we suggest reviewing and optimizing the Entity–Relationship Model (ERM), where redundancies and inefficiencies contrary to good database design practices were detected. A structural review and the normalization of certain tables would reduce ambiguities, improve logical consistency and enable more robust analytical processes.
- Expand the temporal and geographic scope of the analysis: incorporating new stores and more years of information will make it possible to validate the stability of the observed patterns and capture seasonal variations or those associated with the economic context.
- Incorporate additional customer value metrics: integrating indicators such as margin, net profitability or annual recurrence will make it possible to understand the economic contribution of each profile and align loyalty program incentives with the real value of the customer.
- Optimize the technical infrastructure: migrating part of the processing to a cloud or high-performance environment will make it possible to run more complex models (such as variational autoencoders or hierarchical clustering), which are currently limited by memory and processing capacity.
- Develop predictive models: once a more defined and robust segmentation built on clean data is available, the results can be used as input for more complex predictive models, such as churn prediction, personalized recommendations or purchase propensity estimation. This would make it possible to move toward more dynamic and adaptive loyalty strategies, where commercial actions are adjusted based on churn risk, the customer's projected value and their evolution across the different behavioral profiles.
- Deepen the analysis by supplier: connecting the results with commercial agreements and the assortment would make it possible to identify joint opportunities between the chain and its strategic partners (promotions, shared incentives, assortment strategies by segment).
- Formalize an ETL roadmap: defining sources and the cleaning, normalization and periodic update processes in Snowflake will make it possible to recalculate embeddings and clusters in a stable and automated way, integrating their results into the analytical dashboard.

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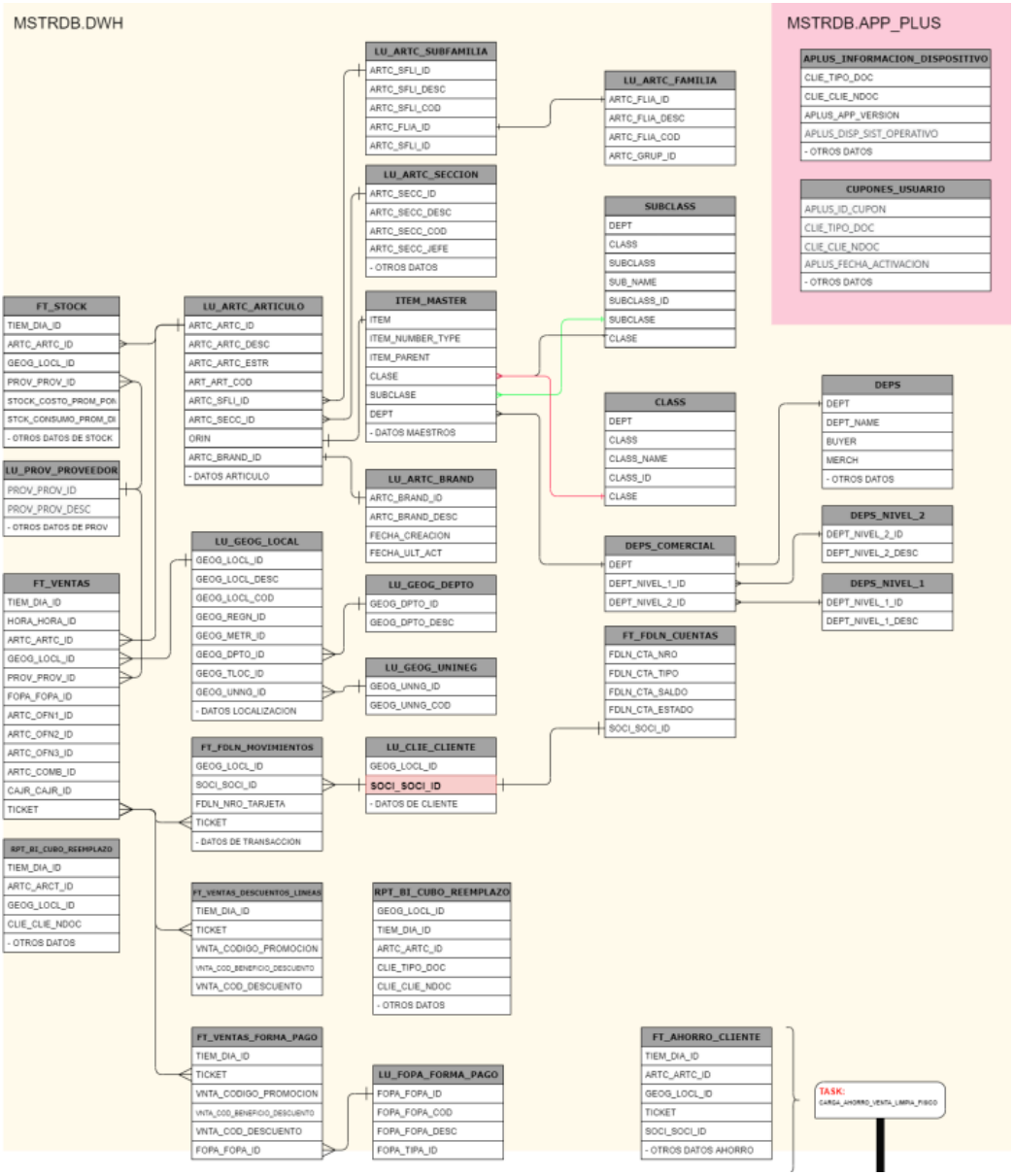
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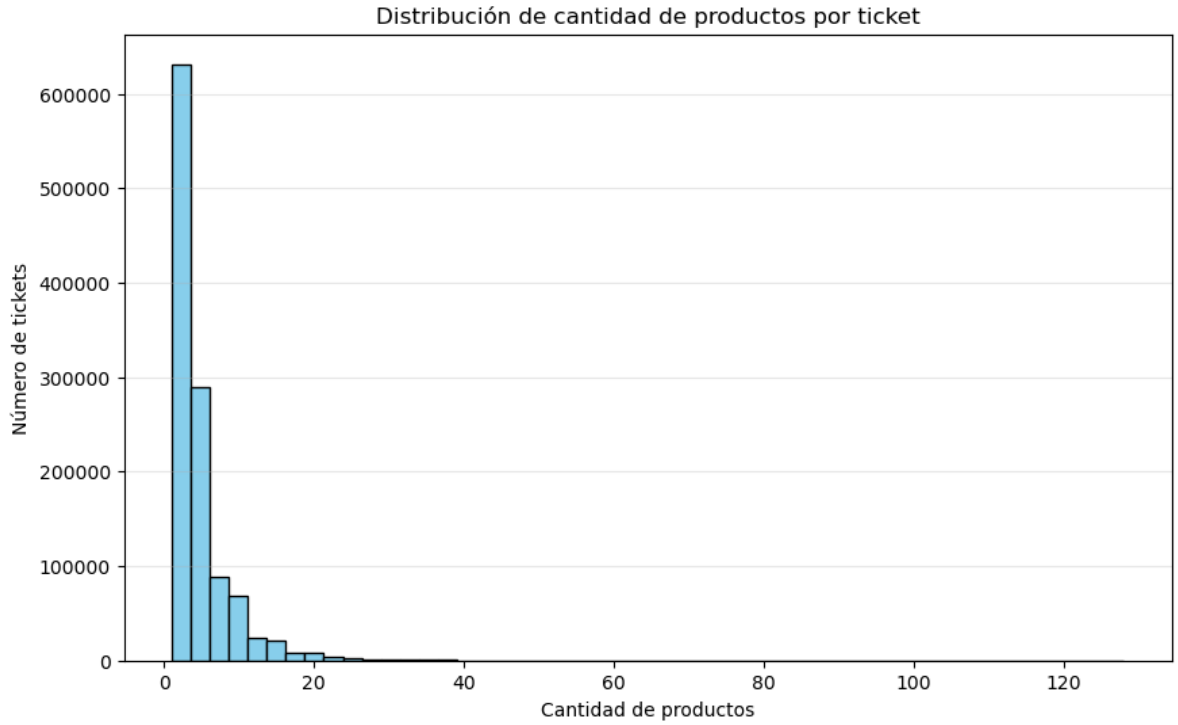
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APPENDIXS

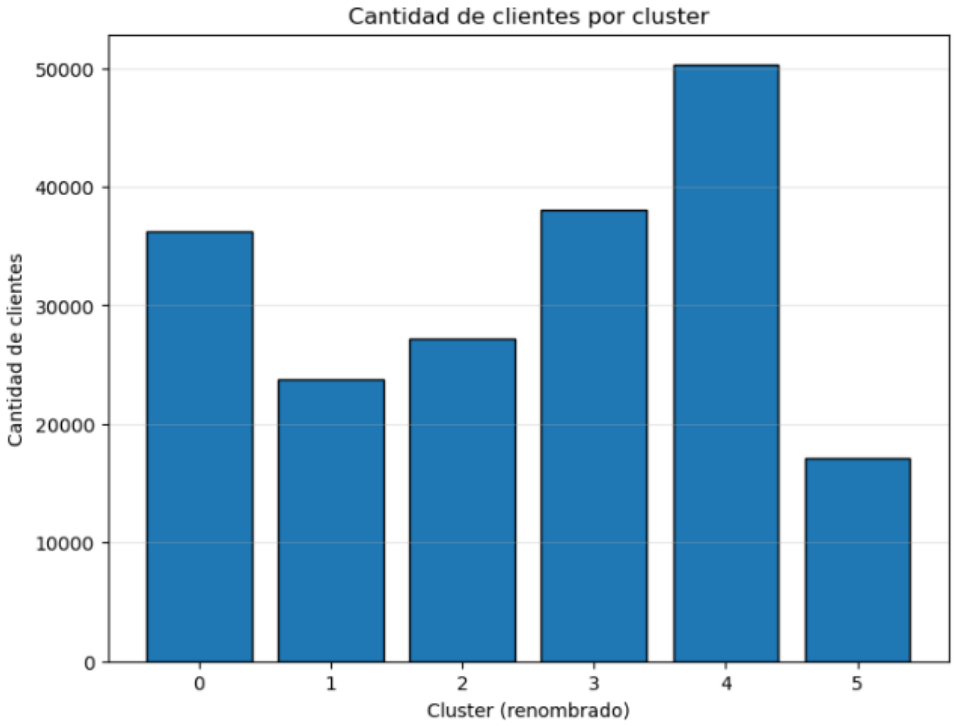
Appendix 1 - Entity-Relationship Model of the database

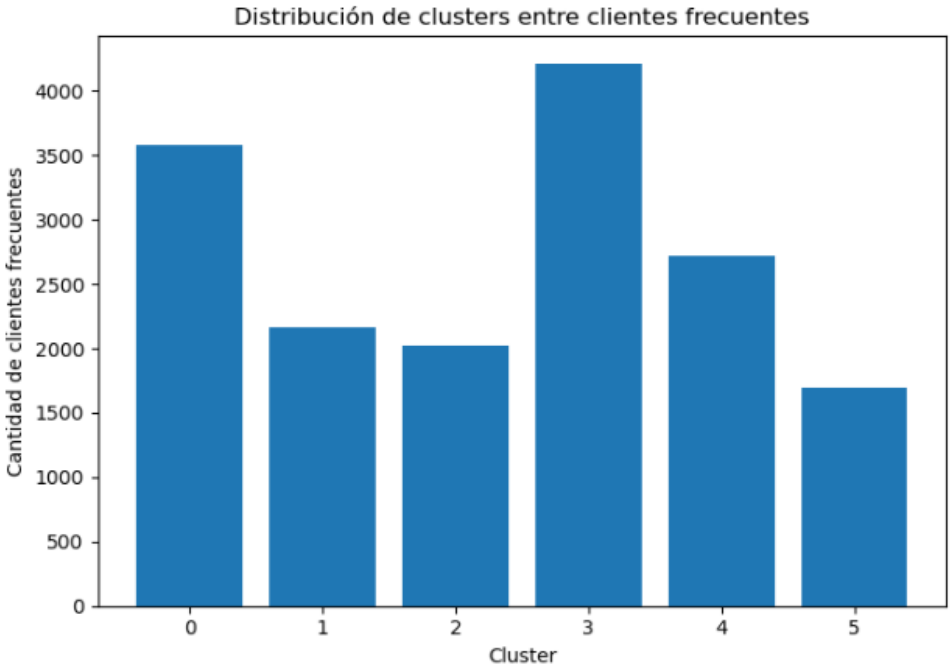


Appendix 2 - Chart used to determine the Word2Vec context window



Appendix 3 - Distribution of frequent and non-frequent customers by cluster





Appendix 4 – Queries used to obtain and transform the PBI input data

---- TRANSACTIONAL TABLE

```
SELECT
    TO_NUMBER(TO_VARCHAR(V.TIEM_DIA_ID, 'YYYYMM'))  AnioMes,
    V.GEOG_LOCL_ID                                  IdSucursal,
    SC.SUBCLASE                                     CodSubClase,
    V.PROV_PROV_ID                                  IdProveedor,
    SUM(V.VNTA_UNIDADES)                            CantidadVendida,
    SUM(V.VNTA_IMPORTE_CON_IVA)                     SumaImporte
FROM MSTRDB.DWH.FT_VENTAS           V
JOIN MSTRDB.DWH.LU_ARTC_ARTICULO    A  ON A.ARTC_ARTC_ID = V.artc_artc_id
JOIN MSTRDB.DWH.ITEM_MASTER         I  ON I.item         = A.orin
JOIN MSTRDB.DWH.CLASS               C  ON C.clase        = I.clase
JOIN MSTRDB.DWH.SUBCLASS            SC ON SC.clase       = I.clase
                                          AND SC.subclase = I.subclase
JOIN MSTRDB.DWH.LU_GEOG_LOCAL       G  ON G.GEOG_LOCL_ID = V.GEOG_LOCL_ID
WHERE TO_NUMBER(TO_VARCHAR(V.TIEM_DIA_ID, 'YYYYMM')) >= (YEAR(CURRENT_DATE())- 2)* 100 + 01 |
    AND C.CLASE NOT IN (1980001, 490002, 910001)
    AND SC.SUBCLASE NOT IN (300010001, 13600020002, 13600010002, -1)
    AND G.GEOG_UNNG_ID NOT IN (3, 4, 5, 7, 8, 9, 11)
    AND G.GEOG_TLOC_ID NOT IN (6, 7, 11, 12)
    AND G.GEOG_LOCL_ID NOT IN (200, 202, 204, 307, 221, 306, 155, 122, 123, 174, 152, 70)
GROUP BY TO_NUMBER(TO_VARCHAR(V.TIEM_DIA_ID, 'YYYYMM')), V.GEOG_LOCL_ID, SC.SUBCLASE , V.PROV_PROV_ID
ORDER BY TO_NUMBER(TO_VARCHAR(V.TIEM_DIA_ID, 'YYYYMM')) ASC, V.GEOG_LOCL_ID, SC.SUBCLASE , V.PROV_PROV_ID
```

--- LOOKUP TABLES

-------- suppliers lookup table

```
SELECT DISTINCT P.PROV_PROV_ID      IdProveedor,
                P.PROV_PROV_COD     CodProveedor,
                P.PROV_PROV_DESC    Proveedor,
                P.PROV_PROV_PAIS    PaisProveedor
FROM MSTRDB.DWH.FT_VENTAS           V
JOIN MSTRDB.DWH.LU_PROV_PROVEEDOR   P  ON P.PROV_PROV_ID = V.PROV_PROV_ID
WHERE TO_NUMBER(TO_VARCHAR(V.TIEM_DIA_ID, 'YYYYMM')) >= TO_NUMBER(TO_VARCHAR(ADD_MONTHS(CURRENT_DATE(), -24), 'YYYYMM'))
```

-------- Subclasses lookup table

```
WITH TipoVentaPorSubclase AS (
    SELECT
        SC.SUBCLASE AS CodSubClase,
        CASE
            WHEN SUM(CASE WHEN MOD(V.VNTA_UNIDADES, 1) > 0 THEN 1 ELSE 0 END) > 0 THEN 'peso'
            ELSE 'unidad'
        END AS TipoVenta
    FROM MSTRDB.DWH.FT_VENTAS V
    JOIN MSTRDB.DWH.LU_ARTC_ARTICULO A ON A.ARTC_ARTC_ID = V.artc_artc_id
    JOIN MSTRDB.DWH.ITEM_MASTER I ON I.item = A.orin
    JOIN MSTRDB.DWH.SUBCLASS SC ON SC.clase = I.clase AND SC.subclase = I.subclase
    WHERE TO_NUMBER(TO_VARCHAR(V.TIEM_DIA_ID, 'YYYYMM')) >= TO_NUMBER(TO_VARCHAR(ADD_MONTHS(CURRENT_DATE(), -36), 'YYYYMM'))
        AND V.VNTA_UNIDADES > 0
        AND V.VNTA_IMPORTE_CON_IVA > 0
        AND SC.SUBCLASE NOT IN (300010001, 13600020002, 13600010002)
    GROUP BY SC.SUBCLASE
)

SELECT DISTINCT
    SC.SUBCLASE       AS CodSubClase,
    SC.SUB_NAME       AS SubClase,
    SC.CLASE          AS CodClase,
    TV.TipoVenta      AS TipoVenta
FROM MSTRDB.DWH.ITEM_MASTER I
JOIN MSTRDB.DWH.SUBCLASS SC ON SC.clase = I.clase AND SC.subclase = I.subclase
INNER JOIN TipoVentaPorSubclase TV ON TV.CodSubClase = SC.SUBCLASE
```

-------- class lookup table

```
SELECT DISTINCT CLASE       CodClase,
                CLASS_NAME  Clase
FROM MSTRDB.DWH.CLASS
```

-------- stores lookup table

```
WITH SucursalesConVentas AS (
    SELECT DISTINCT V.GEOG_LOCL_ID AS IdSucursal
    FROM MSTRDB.DWH.FT_VENTAS V
    WHERE TO_NUMBER(TO_VARCHAR(V.TIEM_DIA_ID, 'YYYYMM')) >= TO_NUMBER(TO_VARCHAR(ADD_MONTHS(CURRENT_DATE(), -36), 'YYYYMM'))
        AND V.VNTA_UNIDADES > 0
        AND V.VNTA_IMPORTE_CON_IVA > 0
)

SELECT DISTINCT
    LOC.GEOG_LOCL_ID       AS IdSucursal,
    LOC.GEOG_LOCL_COD      AS CodSucursal,
    LOC.GEOG_LOCL_DESC     AS Sucursal,
    LOC.GEOG_LOCL_LAT      AS LatitudSucursal,
    LOC.GEOG_LOCL_LON      AS LongitudSucursal,
    LOC.GEOG_LOCL_CIUDAD   AS CiudadSucursal,
    CASE
        WHEN ( LOC.GEOG_LOCL_DESC LIKE '%PEYA%' OR LOC.GEOG_LOCL_DESC LIKE '%ECOMMERCE%' OR LOC.GEOG_LOCL_DESC = 'Darkstore') THEN NULL
        WHEN LOC.GEOG_LOCL_CIUDAD = 'Montevideo' THEN 'Montevideo'
        WHEN (GEOG_LOCL_CIUDAD IS NULL OR GEOG_LOCL_CIUDAD = 'No especificado') AND LOC.GEOG_LOCL_DESC IN ('Shopping', 'Portones') THEN 'Montevideo'
        ELSE 'Interior'
    END                    AS ZonaSucursal,
    CASE
        WHEN ( LOC.GEOG_LOCL_DESC LIKE '%PEYA%' OR LOC.GEOG_LOCL_DESC LIKE '%ECOMMERCE%' OR LOC.GEOG_LOCL_DESC = 'Darkstore') THEN 1
        ELSE 0
    END                    AS VentaDigital,
    CASE WHEN LOC.GEOG_LOCL_DESC LIKE '%PEYA%' THEN 'PEYA' ELSE GTL.GEOG_TLOC_DESC END    AS TipoLocal
FROM MSTRDB.DWH.LU_GEOG_LOCAL LOC
JOIN SucursalesConVentas            S    ON S.IdSucursal = LOC.GEOG_LOCL_ID
JOIN MSTRDB.DWH.LU_GEOG_TIPO_LOCAL  GTL  ON GTL.GEOG_TLOC_ID = LOC.GEOG_TLOC_ID
WHERE LOC.GEOG_UNNG_ID NOT IN (3, 4, 5, 7, 8, 9, 11)
    AND LOC.GEOG_TLOC_ID NOT IN (6, 7, 11, 12)
    AND LOC.GEOG_LOCL_ID NOT IN (200, 202, 204, 307, 221, 306, 155, 122, 123, 174, 152, 70)
```

Appendix 5 – Dashboard measures and parameters

```
_Ventas_Dinamico =
VAR Unidad = SELECTEDVALUE('Unidad_Visualizacion'[Unidad])

RETURN
    SWITCH(TRUE, Unidad = "Unidades", SUM(Trx_Summarized[CANTIDADVENDIDA]), Unidad = "Importe $",SUM
    (Trx_Summarized[SUMAIMPORTE]) )
```

```
_Ventas_Mensual_UltAño =
 VAR MaxMeAnioMes =  CALCULATE(MAX(Trx_Summarized[ANIOMES]), ALL(Trx_Summarized))
VAR AnioMax = LEFT(MaxMeAnioMes,4)
VAR AnioPrevio = AnioMax - 1
VAR MesMax = RIGHT(MaxMeAnioMes,2)


VAR MeAnioMesInicio = VALUE(AnioPrevio & MesMax)

RETURN
    CALCULATE(
    Trx_Summarized[_Ventas_Dinamico],
    Trx_Summarized[ANIOMES] >= MeAnioMesInicio
)
```

```
_Ventas_Año_Parcial =
VAR FechaMaxActual = CALCULATE(MAX(Trx_Summarized[ANIOMES]), ALL(Trx_Summarized))
VAR MesFechaMax = RIGHT(FechaMaxActual,2)
VAR AnioSeleccionado = SELECTEDVALUE(CalendarioMeses[Año])


VAR Inicio = VALUE(AnioSeleccionado & "01")
VAR Corte = VALUE(AnioSeleccionado & MesFechaMax)


RETURN
CALCULATE(
    Trx_Summarized[_Ventas_Dinamico],
    Trx_Summarized[ANIOMES] >= Inicio &&
    Trx_Summarized[ANIOMES] <= Corte
)
```

```
Parametro de Agrupación de Fechas = {
    ("Mensual (Último Año Móvil)", NAMEOF('CalendarioMeses'[Short_Month]), 0),
    ("Quarter", NAMEOF('CalendarioMeses'[AnioQuarter]), 1),
    ("Anual", NAMEOF('CalendarioMeses'[Año]), 2),
    ("Año Parcial", NAMEOF('CalendarioMeses'[Año2]), 3)
}
```

```
_Ventas_Mix =
VAR __SelectedValue =SELECTEDVALUE('Parametro de Agrupación de Fechas'[Parameter Fields])

RETURN
    SWITCH(
    TRUE(),
    __SelectedValue = "'CalendarioMeses'[Short_Month]", CALCULATE(Trx_Summarized[_Ventas_Mensual_UltAño]),
    __SelectedValue = "'CalendarioMeses'[Año2]",         CALCULATE(Trx_Summarized[_Ventas_Año_Parcial]),
                                                            [_Ventas_Dinamico]
    )
```

```
_Ventas_dinamico_PrevYear =
VAR MesesActuales =
    VALUES ( 'CalendarioMeses'[MeAnioMes] )

VAR MesesPrevioAnio =
    SELECTCOLUMNS(
        MesesActuales,
        "AnioMesPrev", [MeAnioMes] - 100
    )

VAR MaxMesReal =
    CALCULATE(
        MAX ( 'Trx_Summarized'[AnioMes] ),
        ALL ( 'Trx_Summarized' )
    )

RETURN
IF( MAX ( 'CalendarioMeses'[MeAnioMes] ) > MaxMesReal, BLANK(),
    CALCULATE(
        [_Ventas_dinamico],
        REMOVEFILTERS ( 'CalendarioMeses' ),
        TREATAS(
            MesesPrevioAnio,
            'Trx_Summarized'[AnioMes]
        )
    )
)
```

```
_Ventas_dinamico_YoY% =
VAR VentasActual    = [_Ventas_dinamico]
VAR VentasPrevYear = [_Ventas_dinamico_PrevYear]
RETURN
IF(
    NOT ISBLANK(VentasPrevYear) && VentasPrevYear <> 0,
    DIVIDE( VentasActual - VentasPrevYear, VentasPrevYear ),
    BLANK()
)
```

```
_Tarj_ParticipacionProveedor =
VAR ImporteFiltro =
    SUM( Trx_Summarized[SUMAIMPORTE] )

VAR ImporteTotal =
    CALCULATE(
        SUM( Trx_Summarized[SUMAIMPORTE] ),
        REMOVEFILTERS( Proveedores )
    )

RETURN
    DIVIDE( ImporteFiltro, ImporteTotal )
```

```
_Tarj_ParticipacionSubclase =
VAR ImporteFiltro =
    SUM( Trx_Summarized[SUMAIMPORTE] )

VAR ImporteTotal =
    CALCULATE(
        SUM( Trx_Summarized[SUMAIMPORTE] ),
        REMOVEFILTERS( Subclases )
    )

RETURN
    DIVIDE( ImporteFiltro, ImporteTotal )
```

```
_Tarj_ParticipacionSucursal =
VAR ImporteFiltro =
    SUM( Trx_Summarized[SUMAIMPORTE] )

VAR ImporteTotal =
    CALCULATE(
        SUM( Trx_Summarized[SUMAIMPORTE] ),
        REMOVEFILTERS( Sucursales )
    )

RETURN
    DIVIDE( ImporteFiltro, ImporteTotal )
```

```
_Tarj_ImporteTotal_MismaAlturaAñoAnterior =
VAR UltimoAnioMes =
    CALCULATE(
        MAX ( Trx_Summarized[ANIOMES] ),
        ALL ( Trx_Summarized[ANIOMES] )
    )

VAR PenultimoAno =  VALUE( LEFT(UltimoAnioMes, 4)) - 1
VAR UltimoMes =  VALUE( RIGHT(UltimoAnioMes, 2))

RETURN
CALCULATE(
    SUM ( Trx_Summarized[SUMAIMPORTE] ),
    CalendarioMeses[Año] = PenultimoAno , CalendarioMeses[MesNro] <= UltimoMes
)
```

STATISTICAL ANALYSIS

To rigorously evaluate the temporal patterns in the supermarket's sales series, a statistical analysis was incorporated to test for the existence of trend, seasonality and weekly seasonality. First, the ADF (Augmented Dickey-Fuller) unit root test was applied to determine whether the series is stationary or shows evidence of a stochastic trend. Then, the trend, seasonal and residual components were explicitly extracted using STL decomposition, making it possible to statistically test whether seasonality differs systematically from zero. Finally, the presence of weekly seasonality was evaluated using a one-way ANOVA, comparing mean sales across days of the week. Together, these tests make it possible to formally characterize the temporal dynamics of the series.

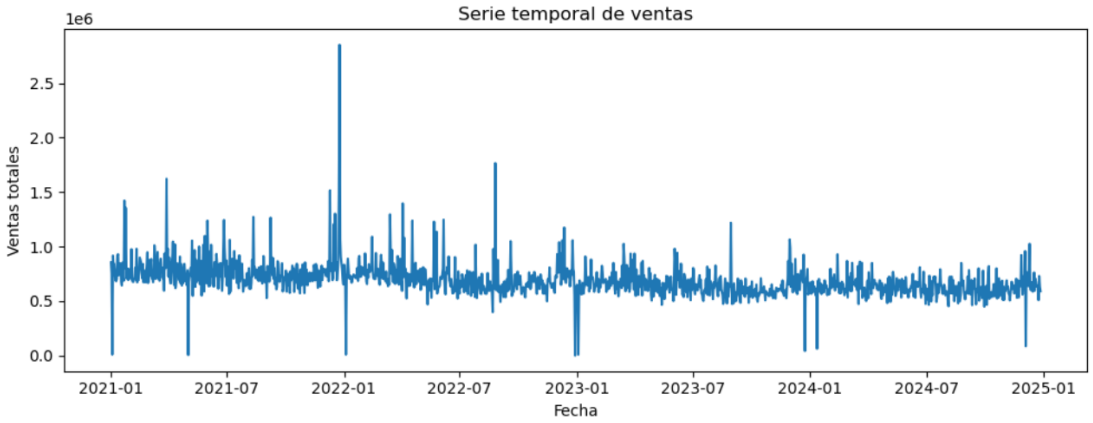


Figure 4.3.3 - Daily sales time series (2021–2025).

The original series shows considerable volatility with pronounced peaks, especially at the beginning. These extreme behaviors suggest the occurrence of exceptional sales events, particularly concentrated in the last days of each year.

The results of the ADF test indicate that the sales series does not have a unit root. The statistic obtained (–4.76) and its associated p-value (0.0000638) make it possible to reject the null hypothesis of non-stationarity at a 95% confidence level. This suggests that the series is stationary in stochastic terms and therefore does not show a persistent trend evolving over time, which is consistent with the temporal structure observed in the data.

The time series decomposition separates the observed data into its fundamental components: trend, seasonality and residual.

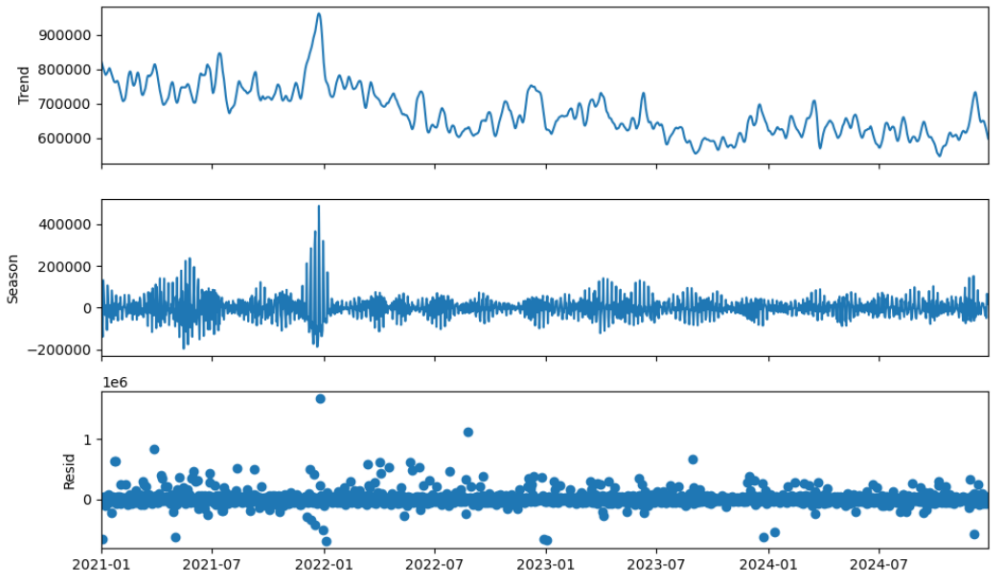


Figure 4.3.4 - Decomposition of the sales series into trend, seasonality and residual.

The long-term trend shows a non-linear evolution of sales. Sustained growth is identified during 2021, reaching a maximum toward the end of that year or the beginning of 2022. From that point on, there is a phase of general contraction during 2022 and 2023, characterized by a gradual decrease in the sales base. Toward 2024, the trend stabilizes at a lower level, which indicates that the decline stopped, although without signs of recovery toward the previous peak values.

Regarding seasonality, the series shows recurring oscillations throughout the period analyzed, which visually suggests the presence of cyclical patterns associated with regular temporal factors, possibly linked to weekly or monthly cycles or to annual holidays that boost demand. In addition, the amplitude of these cycles seems to diminish in the most recent years (2023–2024) compared to 2021–2022, which could reflect a change in the sensitivity of sales to these seasonal factors.

In order to evaluate the presence of systematic seasonality, a t-test was performed to test whether the seasonal component extracted through STL decomposition has a mean different from zero. H0: μ seasonal = 0 was tested. The p-value obtained (0.998) is far above the 5% significance threshold. Therefore, the null hypothesis is not rejected, which indicates that, although there are visible fluctuations over time, they are not consistent or stable enough to be considered a persistent seasonal pattern. Consequently, the statistical evidence suggests that the series does not show systematic seasonality, beyond the occasional variations observed in the chart.

To evaluate the presence of weekly seasonality in the sales series, a one-way ANOVA was applied, using as groups the sales values for each day of the week. The null hypothesis states that mean sales are equal across the seven days, which would imply the absence of systematic differences associated with the day of the week. The p-value obtained (0.602) is considerably higher than the 5% significance level, so the null hypothesis is not rejected. Consequently, there is no statistical evidence of strong weekly seasonality, which suggests that, on average, sales do not differ significantly between the different days of the week.

Finally, the residual component captures the variation not explained by the trend or seasonality, including both noise and random events. The concentration of residuals around zero indicates that the decomposition adequately captures most of the dynamics of the series. However, the presence of

significant outliers reveals the existence of large atypical events, beyond the regular patterns, that impacted sales, most notably the extraordinary peak recorded at the end of 2021.